Anot: Oksitlenmenin gerçekleştiği, metalin elektron vererek iyonlarına ayrıştığı bölgedir.
Katot: İndirgenmenin gerçekleştiği, Anottan gelen elektronların harcandığı bölgedir.
Elektrolit: İyonların taşınmasını sağlayan sıvı ortamdır. Akım metalin içinde e^-lar ile taşınırken elektrolitin içinde iyonlar aracılığıyla taşınır.
Elektriksel yol: Elektronların anottan kataota geçişini sağlayan iletken yoldur.

Metal kaplama işi katotta
Metal kaybı (korozyon) Anotta

- **1 Amper (A)** = 1.000.000 μA (Mikroamper)
- **1 Miliamper (mA)** = 1.000 µA (Mikroamper)
- **1 Mikroamper (µA)** = 0,001 mA (Miliamper)
- **1 Mikroamper (µA)** = 0,000001 A (Amper)

Faraday's laws, the Electrochemical Cell and Electrode Potential

Electron (e^-)

Avogadro's number (N_A) $6{,}022 \times 10^{23}$

Faraday constant (F) $= N_A \times e^- = 96485$ C/mol

Current (I) and current density (i) I (Amper) $i = \frac{I}{A}$ → area

Potential (ε) (volt) $\Delta G = -zFE$

Resistance (R) and resistivity (ρ_s) $R = \frac{V}{I}$

Conductivity (σ) $= \frac{1}{\rho}$ ↗ Özdirenç

chain to remember: $I \cdot t = Q \rightarrow Q/F = \text{mole}^- \rightarrow \div z = \text{mol metal} \rightarrow$

The arithmetic of electricity

charge (Q) (Coulomb) $1C = 1A \times 1S$

(current (I))

Resistance (R) Direnç → Ohm Ω

Potential (V, E)

Energy

Power

Faraday Zaman x Akım → $I \times t = Q$ ↗ yük

yük bölü Faraday Sabiti → mol e^-

From charge to mass

elektronun mol sayısını metalin değerliğine (z) bölerek o mol sayısını buluruz.

$$m = \frac{M \cdot I \cdot t}{z \cdot F} = \frac{M}{z} Q \qquad z = \frac{M}{zF}$$

$$m_{(\text{mass, g})} = \frac{M\,(\text{g/mol}) \times \overbrace{I\,(\text{A}) \times t\,(\text{s})}^{Q\,(\text{C})}}{\underbrace{z}_{\text{electron number}} \times \underbrace{F}_{96485}}$$

z → elektrokimyasal eşdeğer $I \times t$ yani $Q = I$ olunca elektrokimyasal eşdeğer deniyor.

Teorik (Faraday böyle) → %100

gerçekte böyle değil η = akım verimi

$$\eta = \frac{m_{\text{measured}}}{m_{\text{Faraday}}} = \frac{\text{theoric}}{\text{actual}}$$

Derive Faraday's law in 4 steps

Step 1 → $Q = It$ (charge)

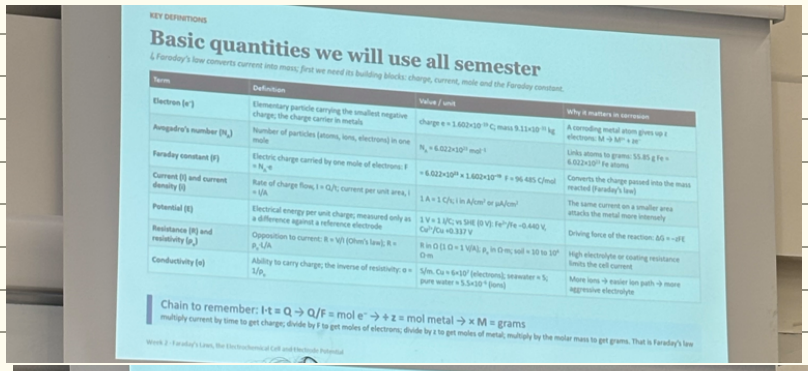
Step 2 → $n_e = \frac{Q}{F} = \frac{It}{F}$ (moles of e^-) * charge → moles of e^- → moles

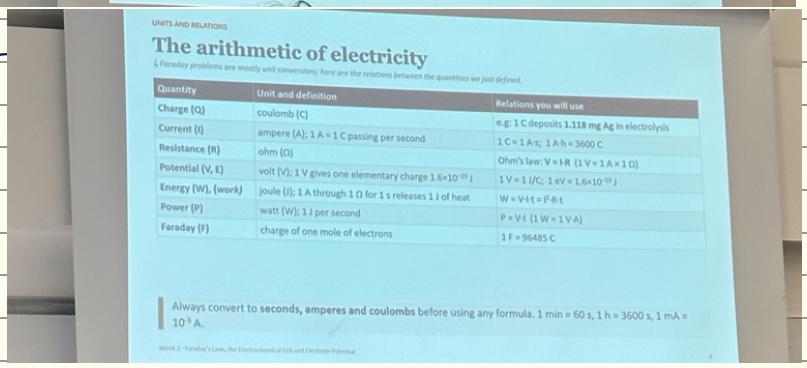
Step 3 → $n_M = \frac{n_e}{z} = \frac{It}{zF}$ (moles of metal)

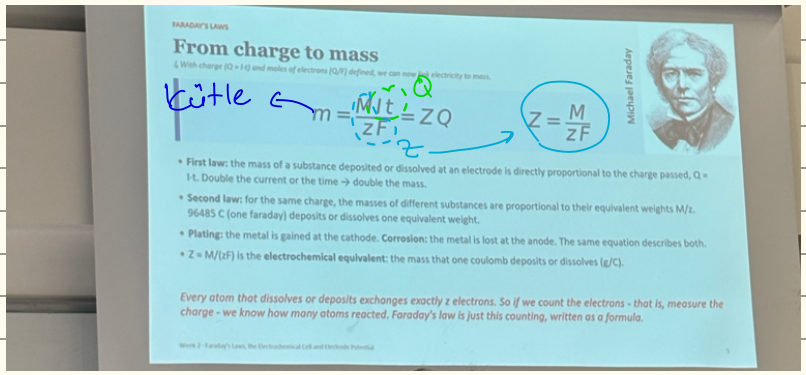
Step 4 → $n_M M = \frac{MIt}{zF}$ (grams)

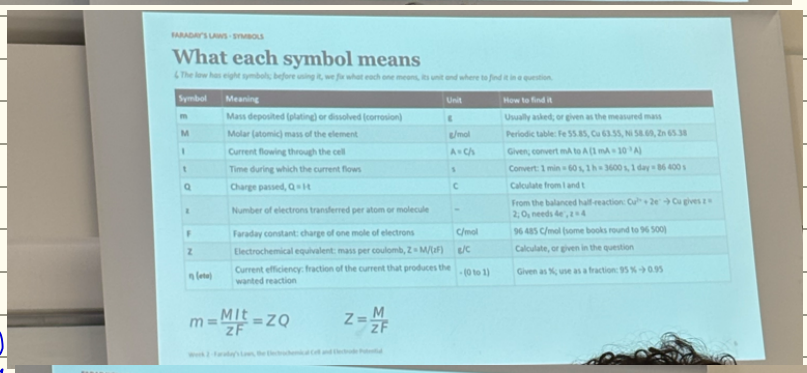
(0-1)

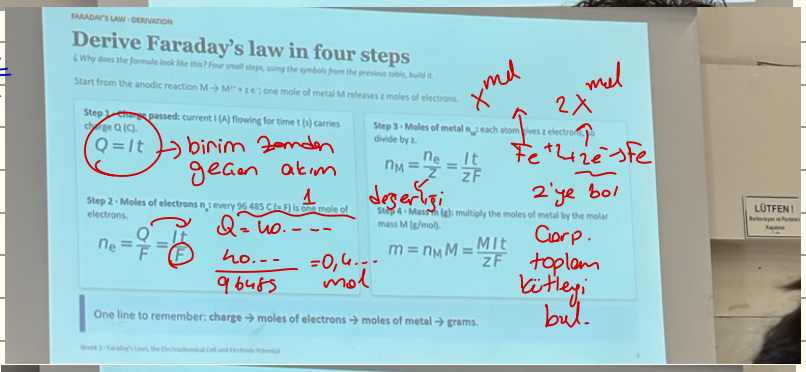
$$\eta = \frac{\downarrow m_{\text{measured}}}{\uparrow m_{\text{Faraday}}} = \frac{\downarrow I_{\text{Theoretical}}}{\uparrow I_{\text{practice}}}$$

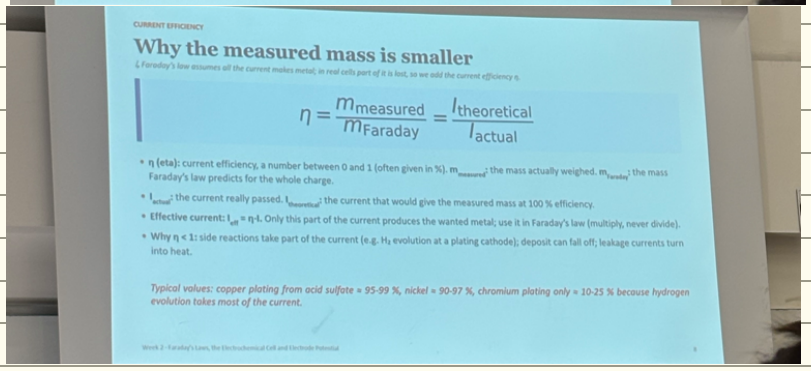






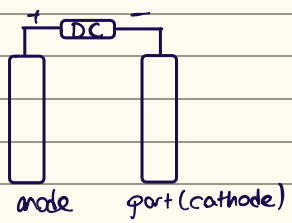






ex

A steel part with a total surface area of 2.5 dm passed for 75 minutes. The current efficiency of the over the whole surface.



area $A = 2.5\ dm^2$

current $I = 1.25\ A$, time

current efficiency $\eta = 9$

$Ni = M = 58.7\ g/mol$, $z =$

$F = 96485\ C/mol$

a) the mass of nickel deposited (g)

b) the coating thickness (μm)

$I_{eff} = \eta I = 0.95 \times 1.25 = 1.1875\ A$

$t = 75 \times 60 = 6500\ s$

$m = \frac{M I_{eff} t}{zF} = \frac{58.7 \times 1.1875 \times 6500}{2 \times 96485} = 1.626\ g$

$x = \frac{m}{\rho} = \frac{1.626}{8.9} = 0.1827\ cm^3$

$A = 2.5\ dm^2 = 250\ cm^3$

$I_{theoretical} = \frac{3600 \cdot 63.5 \cdot 0.026}{+2 \cdot 96485}$

ex

A current of 26 mA is passed through a copper weighed afterwards, 0.0300g of copper has been depos

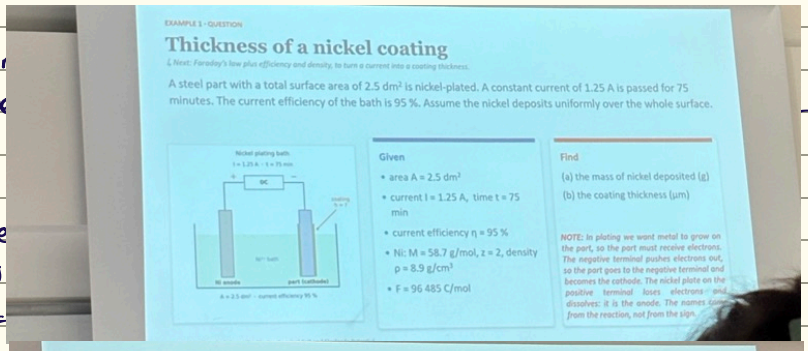
$I = 26\ mA$

$t = 1\ h$

$m = 0.0300\ g$

$Cu = M = 63.5\ g/mol$, $z = 2$

$F = 96485\ C/mol$


EXAMPLE 1 · QUESTION

Thickness of a nickel coating

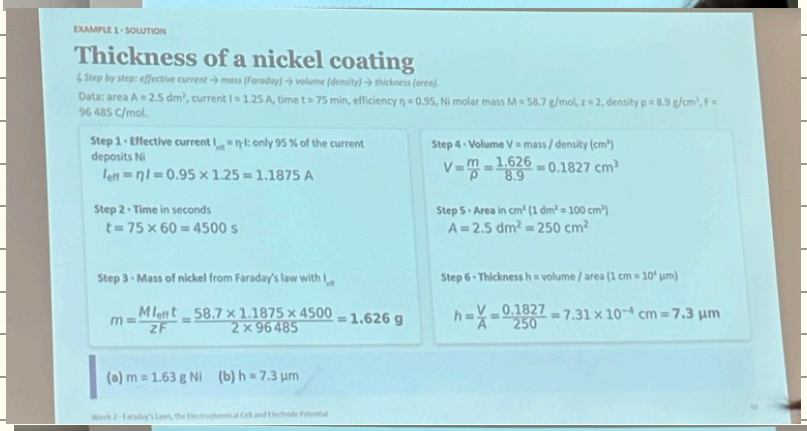
Next: Faraday's law plus efficiency and density, to turn a current into a coating thickness.

A steel part with a total surface area of 2.5 dm² is nickel-plated. A constant current of 1.25 A is passed for 75 minutes. The current efficiency of the bath is 95 %. Assume the nickel deposits uniformly over the whole surface.

Given
- area A = 2.5 dm²
- current I = 1.25 A, time t = 75 min
- current efficiency η = 95 %
- Ni: M = 58.7 g/mol, z = 2, density ρ = 8.9 g/cm³
- F = 96 485 C/mol

Find
(a) the mass of nickel deposited (g)
(b) the coating thickness (μm)

NOTE: In plating we want metal to grow on the part, so the part must receive electrons. The negative terminal pushes electrons out, so the part goes to the negative terminal and becomes the cathode. The nickel plate on the positive terminal loses electrons - and dissolves: it is the anode. The names come from the reaction, not from the sign.


EXAMPLE 1 · SOLUTION

Thickness of a nickel coating

Step by step: effective current → mass (Faraday) → volume (density) → thickness (area).

Data: area A = 2.5 dm², current I = 1.25 A, time t = 75 min, efficiency η = 0.95, Ni molar mass M = 58.7 g/mol, z = 2, density ρ = 8.9 g/cm³, F = 96 485 C/mol.

Step 1 · Effective current $I_{eff} = \eta I$: only 95 % of the current deposits Ni
$I_{eff} = \eta I = 0.95 \times 1.25 = 1.1875\ A$

Step 2 · Time in seconds
$t = 75 \times 60 = 4500\ s$

Step 3 · Mass of nickel from Faraday's law with I_{eff}
$m = \frac{M I_{eff} t}{zF} = \frac{58.7 \times 1.1875 \times 4500}{2 \times 96485} = 1.626\ g$

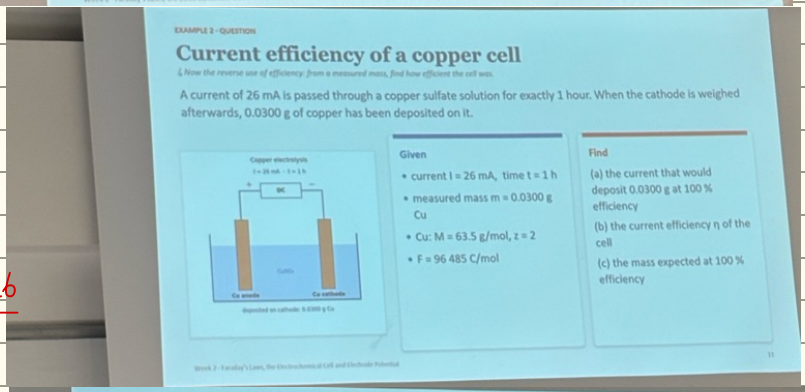
Step 4 · Volume V = mass / density (cm³)
$V = \frac{m}{\rho} = \frac{1.626}{8.9} = 0.1827\ cm^3$

Step 5 · Area in cm² (1 dm² = 100 cm²)
$A = 2.5\ dm^2 = 250\ cm^2$

Step 6 · Thickness h = volume / area (1 cm = 10⁴ μm)
$h = \frac{V}{A} = \frac{0.1827}{250} = 7.31 \times 10^{-4}\ cm = 7.3\ \mu m$

(a) m = 1.63 g Ni (b) h = 7.3 μm

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential


EXAMPLE 2 · QUESTION

Current efficiency of a copper cell

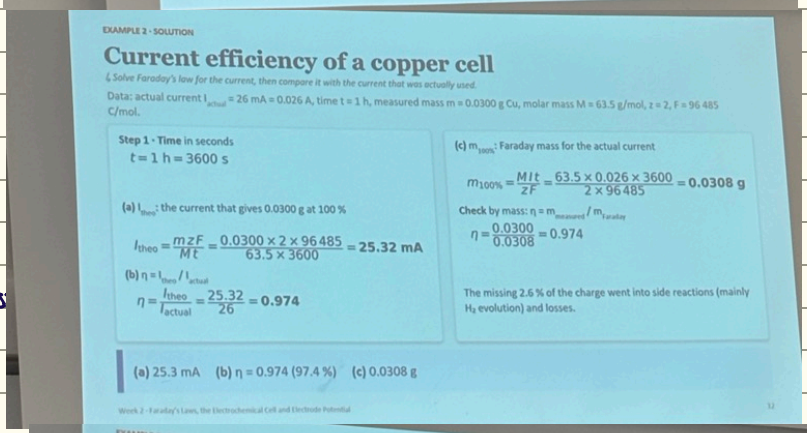
Now the reverse use of efficiency: from a measured mass, find how efficient the cell was.

A current of 26 mA is passed through a copper sulfate solution for exactly 1 hour. When the cathode is weighed afterwards, 0.0300 g of copper has been deposited on it.

Given
- current I = 26 mA, time t = 1 h
- measured mass m = 0.0300 g Cu
- Cu: M = 63.5 g/mol, z = 2
- F = 96 485 C/mol

Find
(a) the current that would deposit 0.0300 g at 100 % efficiency
(b) the current efficiency η of the cell
(c) the mass expected at 100 % efficiency

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential


EXAMPLE 2 · SOLUTION

Current efficiency of a copper cell

Solve Faraday's law for the current, then compare it with the current that was actually used.

Data: actual current I_{actual} = 26 mA = 0.026 A, time t = 1 h, measured mass m = 0.0300 g Cu, molar mass M = 63.5 g/mol, z = 2, F = 96 485 C/mol.

Step 1 · Time in seconds
$t = 1\ h = 3600\ s$

(a) I_{theo}: the current that gives 0.0300 g at 100 %
$I_{theo} = \frac{mzF}{Mt} = \frac{0.0300 \times 2 \times 96485}{63.5 \times 3600} = 25.32\ mA$

(b) $\eta = I_{theo} / I_{actual}$
$\eta = \frac{I_{theo}}{I_{actual}} = \frac{25.32}{26} = 0.974$

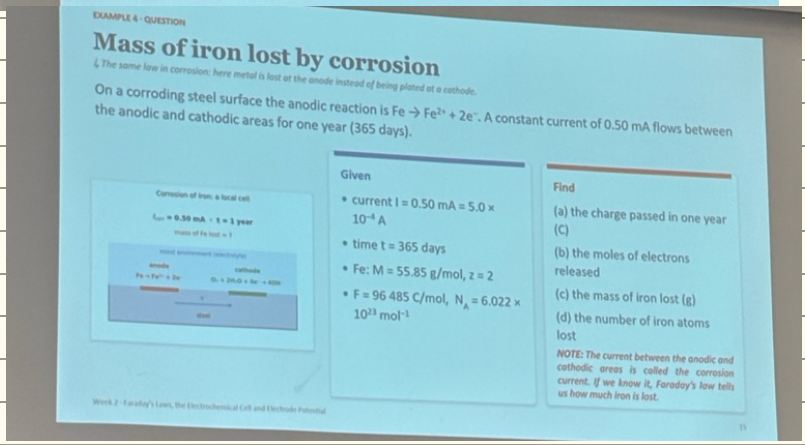
(c) $m_{100\%}$: Faraday mass for the actual current
$m_{100\%} = \frac{MIt}{zF} = \frac{63.5 \times 0.026 \times 3600}{2 \times 96485} = 0.0308\ g$

Check by mass: $\eta = m_{measured} / m_{Faraday}$
$\eta = \frac{0.0300}{0.0308} = 0.974$

The missing 2.6 % of the charge went into side reactions (mainly H_2 evolution) and losses.

(a) 25.3 mA (b) η = 0.974 (97.4 %) (c) 0.0308 g

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential


EXAMPLE 4 · QUESTION

Mass of iron lost by corrosion

The same law in corrosion: here metal is lost at the anode instead of being plated at a cathode.

On a corroding steel surface the anodic reaction is $Fe \rightarrow Fe^{2+} + 2e^-$. A constant current of 0.50 mA flows between the anodic and cathodic areas for one year (365 days).

Given
- current I = 0.50 mA = 5.0 × 10^{-4} A
- time t = 365 days
- Fe: M = 55.85 g/mol, z = 2
- F = 96 485 C/mol, N_A = 6.022 × 10^{23} mol^{-1}

Find
(a) the charge passed in one year (C)
(b) the moles of electrons released
(c) the mass of iron lost (g)
(d) the number of iron atoms lost

NOTE: The current between the anodic and cathodic areas is called the corrosion current. If we know it, Faraday's law tells us how much iron is lost.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

ex

On a corroding steel surface the anodic reaction is $Fe \rightarrow Fe^{2+} + 2e^-$ and the cathodic reaction is $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Assume that 1 mol O_2 reacts for every 2 mol Fe oxidised, and that the current is uniformly distributed over the anodic and cathodic areas for one year (365 days).

a) the charge passed in one year (C)

b) the moles of electrons released

c) the mass of iron lost (g)

d) the number of iron atoms lost

$t = 365 \times 24 \times 3600 = 3.157 \times 10^7$ s

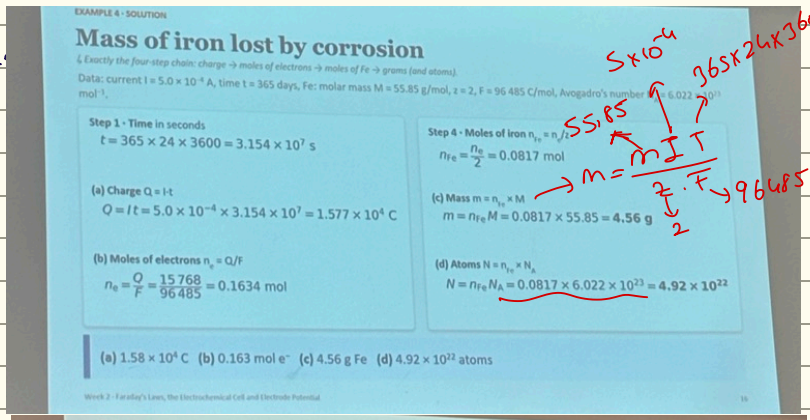
$Q = I t = 5.0 \times 10^{-4} \times 3.154 \times 10^7 = 1.577 \times 10^4$ C

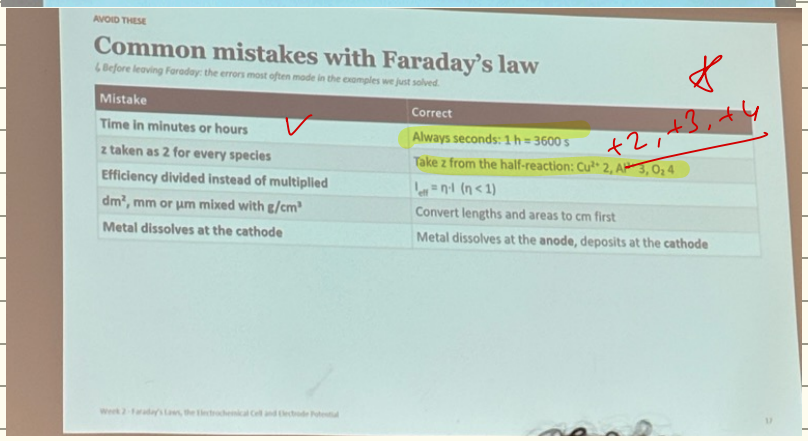
$n_e = \frac{Q}{F} = \frac{15\,768}{96485} = 0.1634$ mol e^-

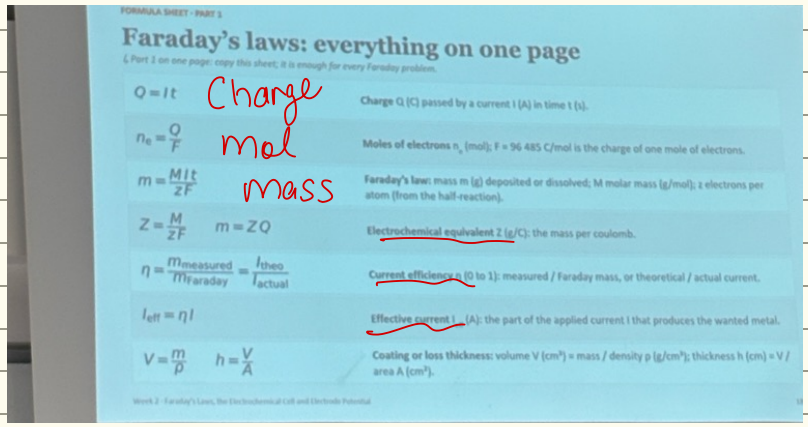
$n_{Fe} = \frac{n_e}{2} = 0.0817$ mol

$m = n_{Fe} M = 0.0817 \times 55.85 = 4.56$ g Fe

$N = n_{Fe} N_A = 0.0817 \times 6.022 \times 10^{23} = 4.92 \times 10^{22}$







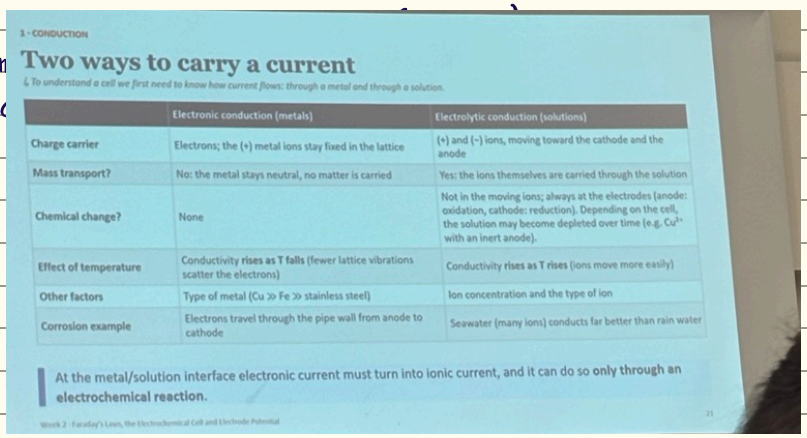
$Q = I t$

$n_e = \frac{Q}{F}$

$z = \frac{n}{2F}$ $\quad m = zQ$

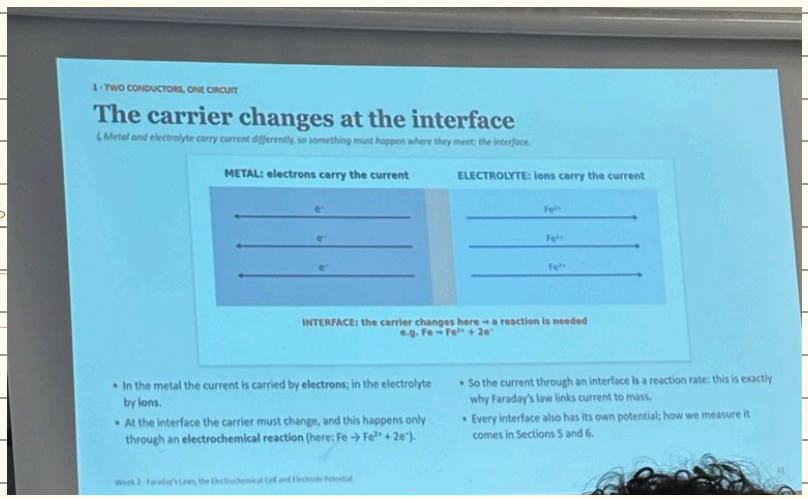
Two ways to carry a current

	electronic conduction (metals)	
Charge carrier	e^-s; the (+) metal ions stay fixed in t	
Mass transport?	no; the metal stays neutral, no matter is	
Chemical change?	none	
Effect of temperature		
Other factors		
Corrosion example		



foto

Is the carrier charges at the interface



metal e^- e^- e^-

electrolyte Fe Fe^{+2} Fe^{+2}

→ In the metal the current is carried by e^-s; in the electrolyte by ions

→ At the interface the carrier

Chemical or electrochemical

Chemical | ec | coupled

No species changes its valence (no electron transferred).

Example: $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$

Fe stays +2, O stays -2

at least one species changes valence: electrons are transferred.

anodic: $Fe \rightarrow Fe^{2+} + 2e^-$

cathodic: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

when the an... clos... , Fe

They

Chemical or electrochemical?

korozyon

Chemical	Electrochemical	Coupled
• No species changes its valence (no electrons transferred).	• At least one species changes valence: electrons are transferred.	• When the anodic and cathodic sites are close, Fe^{2+} and OH^- meet near the surface.
• Example: $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$	• Anodic: $Fe \rightarrow Fe^{2+} + 2e^-$	• They precipitate $Fe(OH)_2$, the first step toward rust.
• Fe stays +2, O stays −2.	• Cathodic: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	

Elementin e^- sayısı değişmez.

e^- transferi olacak

To dissolve in a liquid a metal must first be **oxidized**, so the corrosion of metals is an electrochemical process.

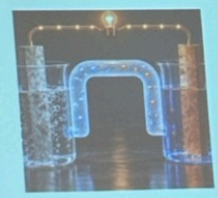
The reaction defines the electrode, not the sign

Cell | Anode | Cathode

Galvanic | − | +

Electrolytic | + | −

The reaction defines the electrode, not the sign



Cell	Anode (oxidation)	Cathode (reduction)
Galvanic (battery in use, corrosion)	Negative (−): source of electrons	Positive (+)
Electrolytic (electrolysis, plating)	Positive (+): connected to the + terminal of the supply	Negative (−)

Anode = where oxidation occurs (electrons are released). **Cathode** = where reduction occurs (electrons are consumed). This is always true; only the **sign** of the anode depends on the type of cell. Example: $Al \rightarrow Al^{3+} + 3e^-$ is anodic, $Al^{3+} + 3e^- \rightarrow Al$ is cathodic.

For basic components

electrolyte | anode | cathode | electrical path

Four basic components

Electrolyte	Anode	Cathode	Electrical path
• A conductive liquid or moist soil that carries ions.	• The electrode where oxidation (metal dissolution) occurs.	• The electrode where reduction occurs (e.g. O_2 or H^+ reduction, metal deposition).	• A metallic connection that carries electrons from anode to cathode; in electrolysis it includes an external power supply.

oksidasyon

redüksiyon

These are the four conditions of corrosion. Remove any one of them and the cell stops; weaken any one of them and corrosion slows down. In spontaneous corrosion the driving force is the anode-cathode **potential difference** and the electrical path is the metal itself.

Galvanic, electrolytic, concentration

Even one metal can make a cell

Type	Characteristic	Example
Galvanic		
Electrolytic		
Concentration		

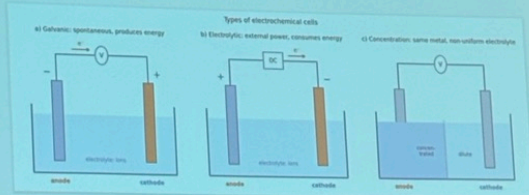
Galvanic

Reaksiyon kendiliğinden Gerçekleşir Anot − katot + enerji üretilir

Electrolytic

Enerji harcanır Anot + katot −

Galvanic, electrolytic, concentration



- **Galvanic:** the reaction is spontaneous and produces electrical energy (power P = E·I). Anode (−), cathode (+).
- **Electrolytic:** an external power supply forces a non-spontaneous reaction; it consumes energy. Anode (+), cathode (−).
- **Concentration:** the same metal on both sides, but the electrolyte is not uniform (different concentration or oxygen content).
- In an electrolytic cell the applied voltage must be larger than the galvanic voltage of the reverse reaction.

Open Circuit: The cell corrodes itself.

Open c. | Closed c. | Cell drag.

Where does the potential come from?

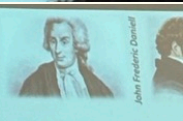
Sum of interfaces | Standard and no...

A cell voltage is the sum of the potential differences at its interfaces: two electrode/electrolyte interfaces, plus ionic contact and liq.-junction potentials

The largest contributions

Zn daha aktif o yüzden e^- verir Anottur. Bakır soy metal e^- alır. katot.

A galvanic cell, closed circuit



- Zn in $ZnSO_4$ is the **anode**: $Zn \rightarrow Zn^{2+} + 2e^-$. (*Zn is more active (-0.763 V); it undergoes oxidation.*)
- Cu in $CuSO_4$ is the **cathode**: $Cu^{2+} + 2e^- \rightarrow Cu$. (*Copper is more noble (+0.337 V) and gets reduced.*)
- Electrons flow through the wire from Zn to Cu.
- In the solution cations move toward Cu and SO_4^{2-} toward Zn; in the salt bridge K^+ moves toward Cu and Cl^- toward Zn.
- The salt bridge (agar + KCl or KNO_3) lets ions pass and keeps both sides electrically neutral.
- Without the bridge, charge builds up and the reaction stops within moments.
- $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.337 - (-0.763) = +1.100$ V

(25 °C, a = 1 (standard condition)) *This is the magnitude of the force pushing electrons from zinc to copper: The greater the difference between the two metals' "tendency to give up electrons," the greater the voltage.*

Why a metal surface carries a potential

① when a metal is put into a solution, some atoms leave as ions ($M \rightarrow M^{n+} + ne^-$). Their e^- stay behind.

gerek yok dedi Keep the word interface for us

The potential # like

e^- Anottan katotta

$\Delta G^\circ = -zFE^\circ$

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

ex Converting between references scale

cobalt tahtada

* Standard reduction potentials

Each value is the pot of a metal in a solution of its own ions at unit activity, 25°C, measured against the SHE

⋮

$\Delta G < 0$ (spontaneous)

$\Delta G = -nFE^\circ_{cell}$

n = # of e^- ; F = Faraday constant (96485 C)

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

Two formulas for any cell

$\Delta G^\circ = -nFE^\circ_{cell}$

ex Two cell with iron: Fe-Cu and Fe-Zn

ex * Electrowinning of Copper

Copper is recovered from an acidic $CuSO_4$ solution by electrolysis between an inert anode at the anode. The cell runs at 200 A.

Cu^{2+}/Cu : $E^\circ = +0.337$ V

O_2/H_2O : $E^\circ = 1.229$ V ($O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$)

Current $I = 200$ A ; Cu : $M = 63.55$ g/mol , $z = 2$

$F = 96485$ C / mol

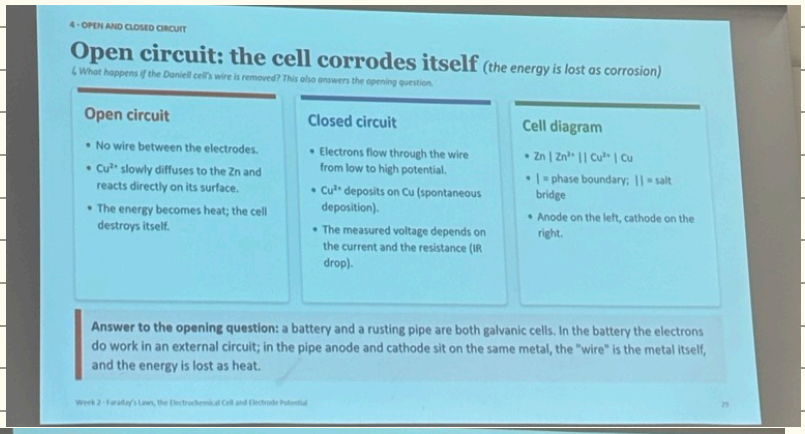
Find

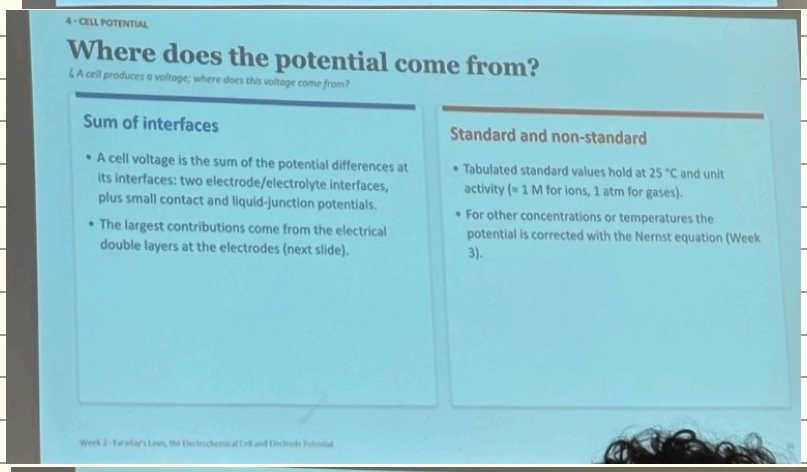
a) the overall cell r

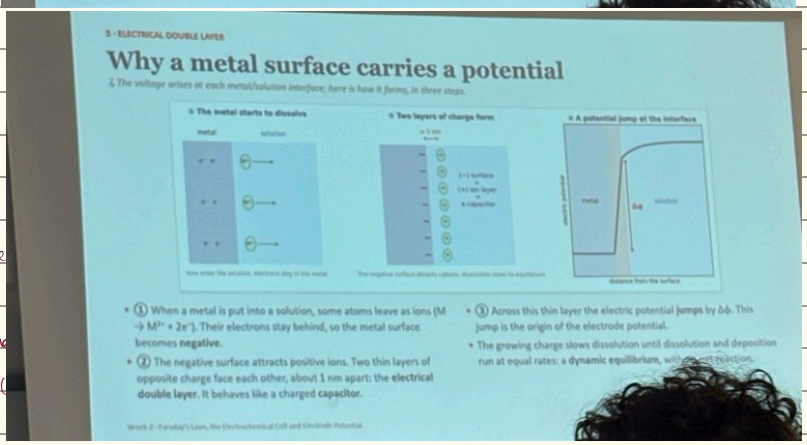
b) the spontaneous

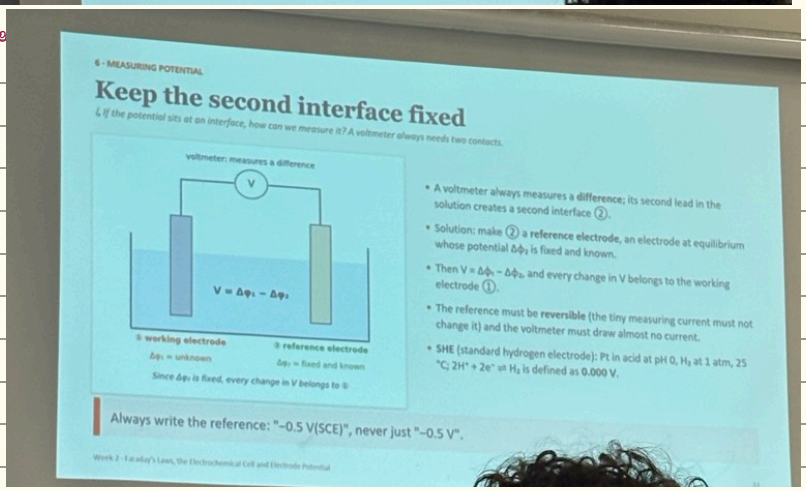
c) the min. voltage

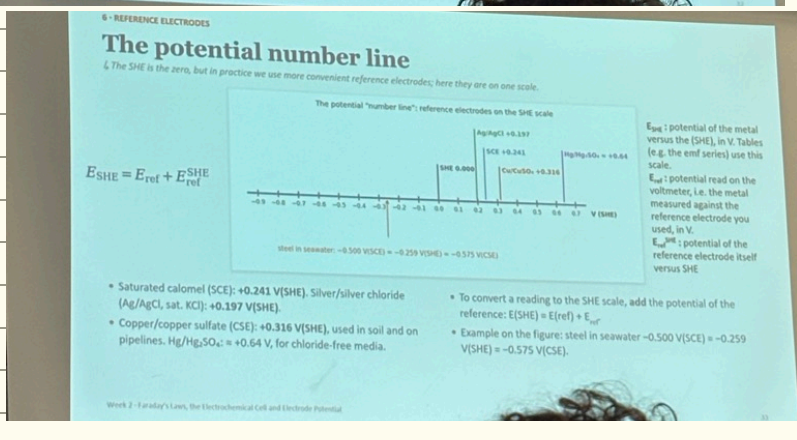
d) the mass of copper

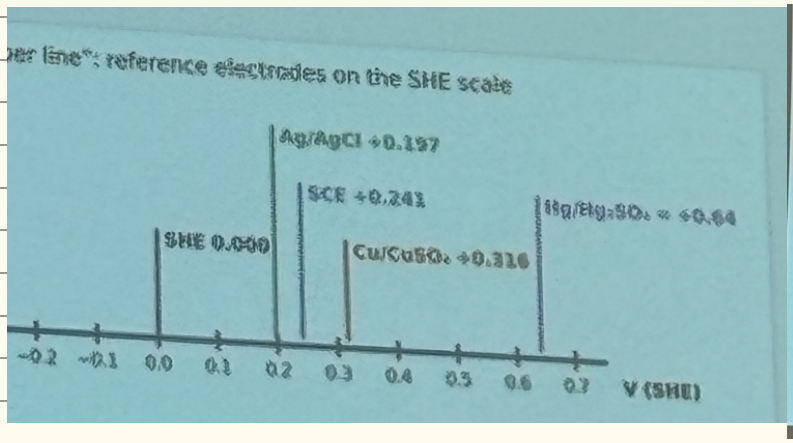








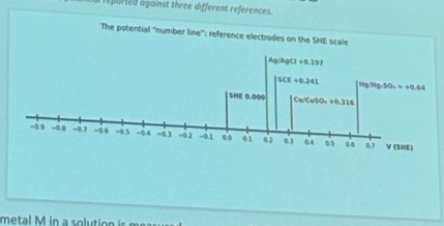




EXAMPLE 5 · QUESTION

Converting between reference scales

Practice with the number line: the same potential reported against three different references.



- The potential of a metal M in a solution is measured against a saturated calomel electrode and reads **−0.541 V(SCE)**.
- **Given:** reference potentials vs SHE at 25 °C: SCE +0.241 V; Ag/AgCl (sat. KCl) +0.197 V; $Cu/CuSO_4$ (CSE) +0.316 V
- **Find:** the potential of M on (a) the SHE scale, (b) the Ag/AgCl scale, (c) the CSE scale
- **Hint:** first place −0.541 V(SCE) on the number line above

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

7 · THE EMF SERIES

Standard reduction potentials

With all potentials on one (SHE) scale, metals can be ranked: this ranking is the emf series.

- Each value is the potential of a metal in a solution of its own ions at unit activity, 25 °C, measured against the SHE (0 V).
- All values are written as **reductions** ($M^{n+} + ze^- \rightarrow M$).
- **Lower (more negative) E° = more active**: this metal tends to oxidize and is the **anode** in a pair.
- **Higher (more positive) E° = more noble**: this metal is the **cathode**.
- Example: Cu +0.337, Fe −0.440, Zn −0.763 V. Against Cu, Fe is the anode; against Zn, Fe is the cathode.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

7 · USING THE EMF SERIES

Two formulas for any cell

The emf series gives two numbers for any cell: its voltage E°_{cell} and its driving force ΔG°.

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

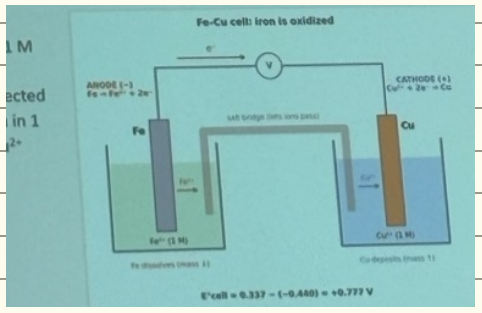
Standard cell potential E°_{cell} (V). $E^\circ_{cathode}$: reduction potential of the cathode (the higher one); E°_{anode}: reduction potential of the anode (the lower one), both taken **directly from the table** without changing signs. $E^\circ_{cell} > 0$ means the cell reaction is spontaneous.

$$\Delta G^\circ = -zFE^\circ_{cell}$$

Standard free-energy change ΔG° (J/mol). z: electrons transferred in the cell reaction; F = 96 485 C/mol. A positive E°_{cell} gives a negative ΔG°: the reaction runs by itself (details in Week 3).

Daniell cell: $E^\circ_{cell} = 0.337 - (-0.763) = +1.100$ V > 0 (spontaneous); $\Delta G^\circ = -212$ kJ/mol

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

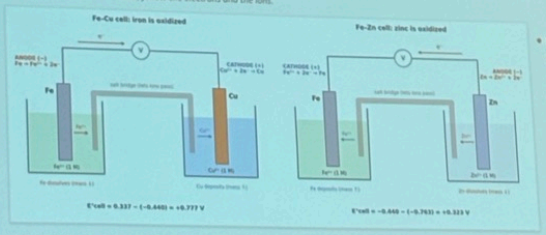


EXAMPLE 6

Two cells with iron: Fe-Cu and Fe-Zn

Step 1: in each pair the lower E° is the anode; follow the electrons and the ions.

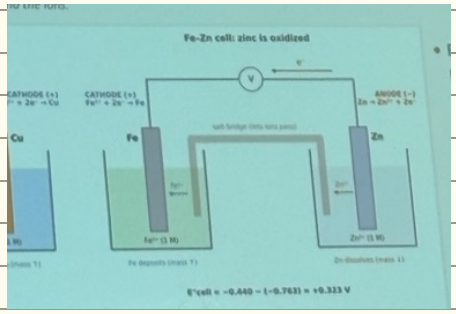
- Fe in 1 M Fe^{2+} connected to Cu in 1 M Cu^{2+}



- Fe in 1 M Fe^{2+} connected to Zn in 1 M Zn^{2+}

- **Given:** 25 °C; E°: Cu^{2+}/Cu +0.337 V, Fe^{2+}/Fe −0.440 V, Zn^{2+}/Zn −0.763 V; F = 96485 C/mol
- **Find:** (a) anode and cathode; (b) half-reactions; (c) E°_{cell}; (d) ΔG°; (e) is iron corroded or protected?
- **Fe-Cu:** $E^\circ(Fe) = -0.440 < E^\circ(Cu) = +0.337$, so iron is the **anode** and dissolves; copper deposits.
- **Fe-Zn:** $E^\circ(Zn) = -0.763 < E^\circ(Fe) = -0.440$, so zinc is the anode; iron is the **cathode** and is protected.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential



EXAMPLE 6 · SOLUTION

Two cells with iron: E° and ΔG°

Step 2: the two formulas from "Using the emf series" give E°_{cell} and ΔG° for each cell.

Data: $E^\circ(Cu^{2+}/Cu) = +0.337$ V, $E^\circ(Fe^{2+}/Fe) = -0.440$ V, $E^\circ(Zn^{2+}/Zn) = -0.763$ V; $z = 2$ electrons; $F = 96\,485$ C/mol.

Cell 1 · (b) anode (oxidation) | cathode (reduction)

$$Fe \rightarrow Fe^{2+} + 2e^- \qquad Cu^{2+} + 2e^- \rightarrow Cu$$

Cell 1 · (c) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$$E^\circ_{cell} = 0.337 - (-0.440) = +0.777 \text{ V}$$

Cell 1 · (d) $\Delta G^\circ = -zFE^\circ_{cell}$

$$\Delta G^\circ = -2 \times 96\,485 \times 0.777 = -150 \text{ kJ/mol}$$

Cell 2 · (b) anode (oxidation) | cathode (reduction)

$$Zn \rightarrow Zn^{2+} + 2e^- \qquad Fe^{2+} + 2e^- \rightarrow Fe$$

Cell 2 · (c) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$$E^\circ_{cell} = -0.440 - (-0.763) = +0.323 \text{ V}$$

Cell 2 · (d) $\Delta G^\circ = -zFE^\circ_{cell}$

$$\Delta G^\circ = -2 \times 96\,485 \times 0.323 = -62.3 \text{ kJ/mol}$$

(e) Fe-Cu: iron **corrodes** (a scratch in a Cu coating pits the steel). Fe-Zn: iron is **protected** (principle of galvanizing).

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

Potansiyeller katsayılarla çarpılmaz.

A+B=C $\in$ =X → 2A+2B+2C $\in$ =X

$Cu^{+2} + 2e^- \rightarrow Cu$ (x2) → $4e^-$

e^- sayısını eşitlemek için "Cu" reaksiyonunu 2 ile çarpıyoruz.

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

$\rightarrow 2Cu^{+2} + 2H_2O \rightarrow 2Cu + O_2 + 4H^+$ (a)

(DC)

Eğer dışarıdan elektrik vermeseydik ve sistem kendi kendine (galvanik) çalışsaydı potansiyeli büyük olan değişen reaksiyonu (+1.229V) katot, düşük olan bakır (+0.337V) anot olurdu.

(b) $E^\circ_{cell} = E^\circ_{katot} - E^\circ_{anot} \Rightarrow E^\circ_{cell} = 1.229 - 0.337 = +0.892\ V$ (spontaneous)

(c) Bizi bakırı çözmek için değil kazanmak istediğimiz için Sistemi tersine çalışmaya zorluyoruz (elektroliz). Bakırı katot, oksijeni anot yapıyoruz. Hücre potansiyeli negatif olur. (0.337 - 1.229 = −0.892V). Bu negatif potansiyeli yenip reaksiyonu başlatabilmek için dışarıdaki DC güç kaynağının sisteme en az +0.892V gerilim uygulanmalıdır.

+0,892 V min Voltage.

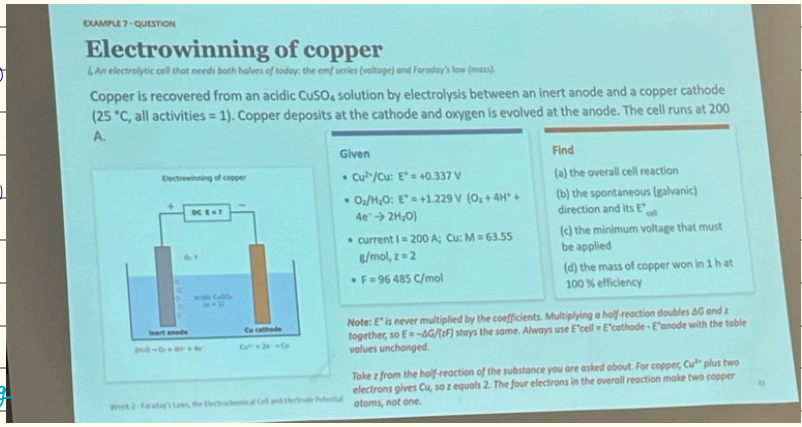
(D) Faraday kanununu kullanarak kütleyi buluruz. (%100 efficiency)

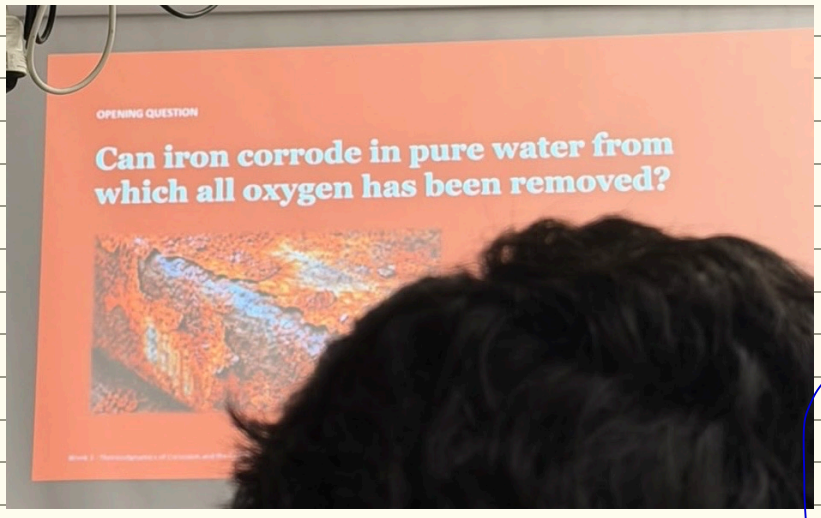
$I = 200A$, $t = 3600s$, $M = 63.55\ g/mol$, $F = 96485\ C/mol$

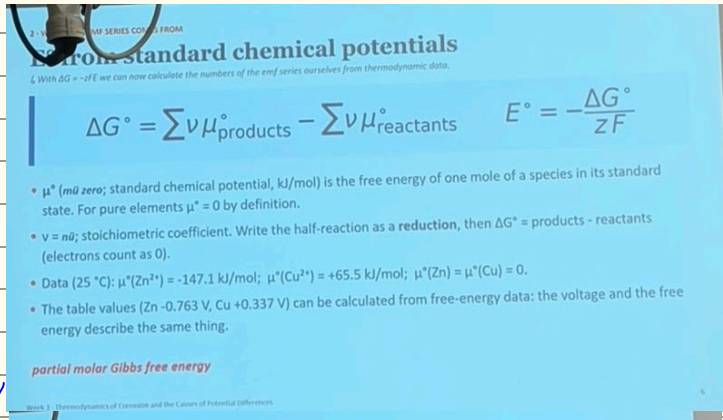
Toplam reaksiyonda $4e^-$ görünse de Cu iyonu (Cu^{+2}) metalik bakıra dönüşürken $2e^-$ alır → $z = 2$

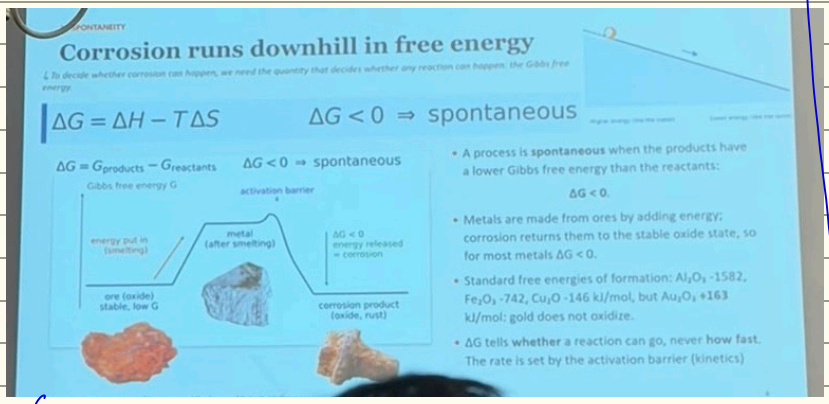
$$m = \frac{MIt}{zF} = \frac{63.55 \times 200 \times 3600}{2 \times 96485} = 237.11g$$

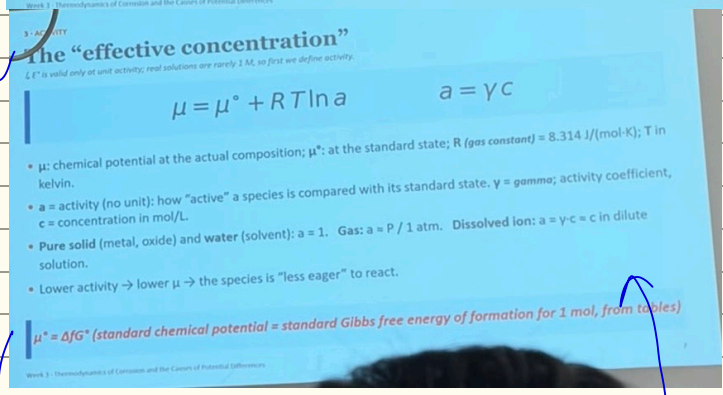
Sistem %100 verimle çalıştığında 1 saatte katottan 237.11 gram bakır birikir.

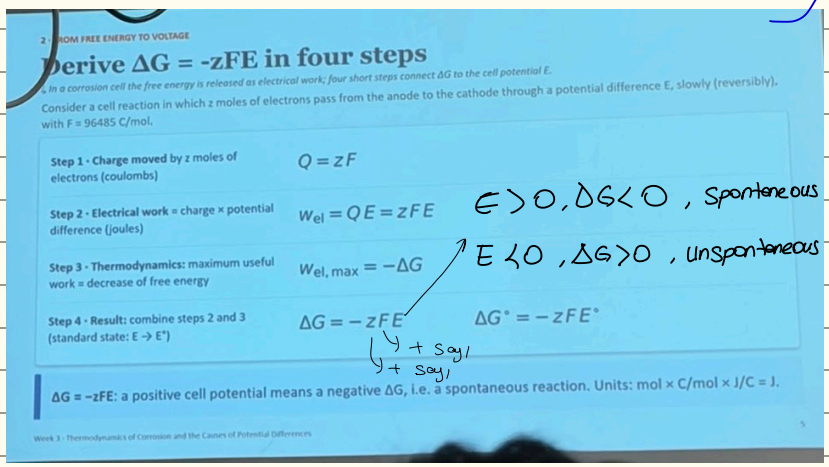








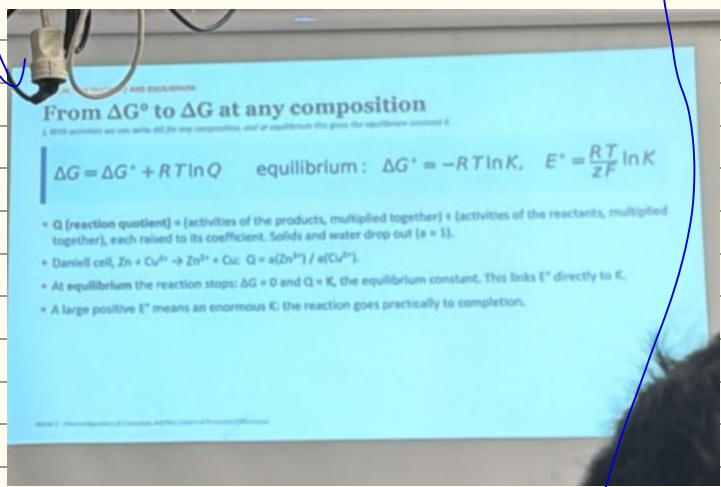




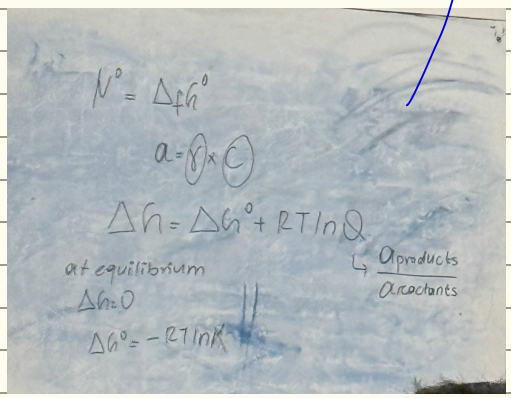
$E > 0, \Delta G < 0$, spontaneous

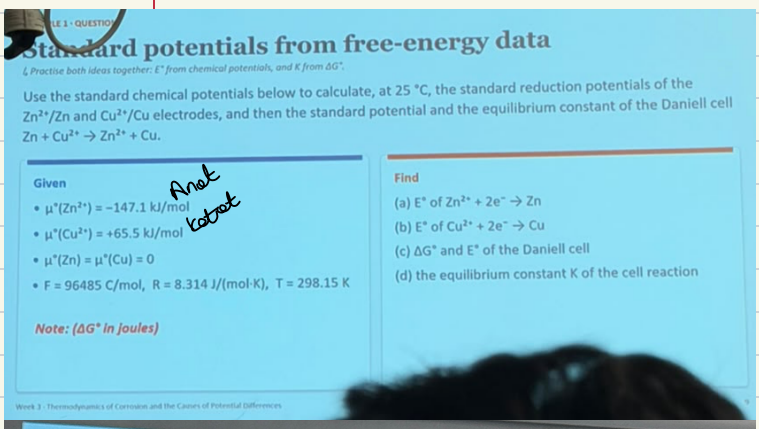
$E < 0, \Delta G > 0$, unspontaneous

(+ sayı, + sayı)



$$E^\circ = \frac{RT}{zF} \cdot \ln K \qquad -\Delta G^\circ$$





Standard potentials from free-energy data

Practise both ideas together: E° from chemical potentials, and K from ΔG°.

Use the standard chemical potentials below to calculate, at 25 °C, the standard reduction potentials of the Zn^{2+}/Zn and Cu^{2+}/Cu electrodes, and then the standard potential and the equilibrium constant of the Daniell cell $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$.

Given

- $\mu^\circ(Zn^{2+}) = -147.1$ kJ/mol
- $\mu^\circ(Cu^{2+}) = +65.5$ kJ/mol
- $\mu^\circ(Zn) = \mu^\circ(Cu) = 0$
- F = 96485 C/mol, R = 8.314 J/(mol·K), T = 298.15 K

Note: (ΔG° in joules)

Find

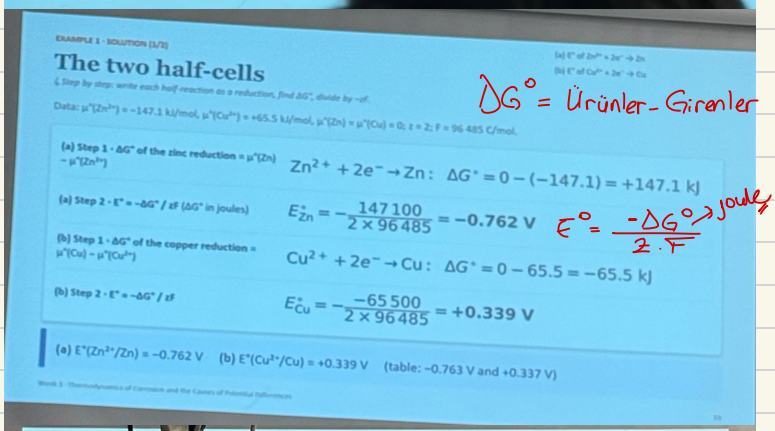
(a) E° of $Zn^{2+} + 2e^- \rightarrow Zn$

(b) E° of $Cu^{2+} + 2e^- \rightarrow Cu$

(c) ΔG° and E° of the Daniell cell

(d) the equilibrium constant K of the cell reaction

Week 3 · Thermodynamics of Corrosion and the Causes of Potential Differences



EXAMPLE 1 · SOLUTION (1/2)

The two half-cells

Step by step: write each half-reaction as a reduction, find ΔG°, divide by −zF.

Data: $\mu^\circ(Zn^{2+}) = -147.1$ kJ/mol, $\mu^\circ(Cu^{2+}) = +65.5$ kJ/mol, $\mu^\circ(Zn) = \mu^\circ(Cu) = 0$; z = 2; F = 96 485 C/mol.

(a) Step 1 · ΔG° of the zinc reduction = $\mu^\circ(Zn) - \mu^\circ(Zn^{2+})$

$$Zn^{2+} + 2e^- \rightarrow Zn: \quad \Delta G^\circ = 0 - (-147.1) = +147.1 \text{ kJ}$$

(a) Step 2 · $E^\circ = -\Delta G^\circ / zF$ (ΔG° in joules)

$$E^\circ_{Zn} = -\frac{147\,100}{2 \times 96\,485} = -0.762 \text{ V}$$

(b) Step 1 · ΔG° of the copper reduction = $\mu^\circ(Cu) - \mu^\circ(Cu^{2+})$

$$Cu^{2+} + 2e^- \rightarrow Cu: \quad \Delta G^\circ = 0 - 65.5 = -65.5 \text{ kJ}$$

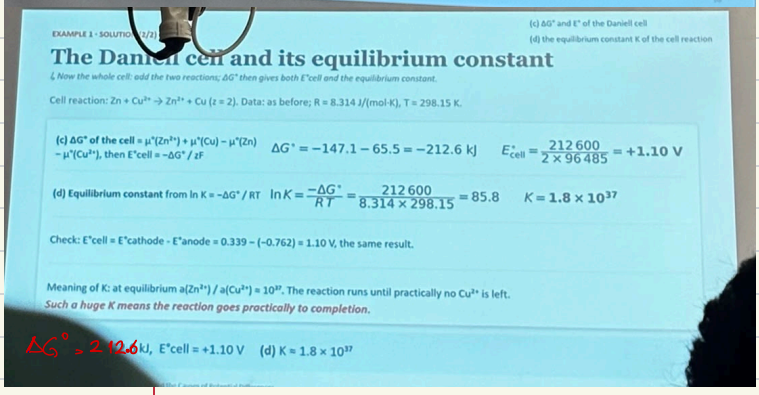
(b) Step 2 · $E^\circ = -\Delta G^\circ / zF$

$$E^\circ_{Cu} = -\frac{-65\,500}{2 \times 96\,485} = +0.339 \text{ V}$$

(a) $E^\circ(Zn^{2+}/Zn) = -0.762$ V (b) $E^\circ(Cu^{2+}/Cu) = +0.339$ V (table: −0.763 V and +0.337 V)

$\Delta G^\circ = \Delta G^\circ_{ürünler} - \Delta G^\circ_{girenler}$

$E^\circ = \frac{-\Delta G^\circ}{zF}$ → joule



EXAMPLE 1 · SOLUTION (2/2)

The Daniell cell and its equilibrium constant

Now the whole cell: add the two reactions; ΔG° then gives both E°cell and the equilibrium constant.

Cell reaction: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ (z = 2). Data: as before; R = 8.314 J/(mol·K), T = 298.15 K.

(c) ΔG° of the cell = $\mu^\circ(Zn^{2+}) + \mu^\circ(Cu) - \mu^\circ(Zn) - \mu^\circ(Cu^{2+})$, then $E^\circ_{cell} = -\Delta G^\circ / zF$

$$\Delta G^\circ = -147.1 - 65.5 = -212.6 \text{ kJ} \qquad E^\circ_{cell} = \frac{212\,600}{2 \times 96\,485} = +1.10 \text{ V}$$

(d) Equilibrium constant from $\ln K = -\Delta G^\circ / RT$

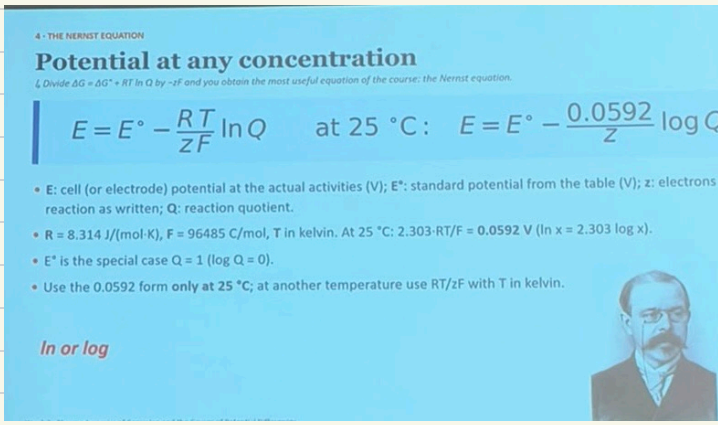
$$\ln K = \frac{-\Delta G^\circ}{RT} = \frac{212\,600}{8.314 \times 298.15} = 85.8 \qquad K = 1.8 \times 10^{37}$$

Check: $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = 0.339 - (-0.762) = 1.10$ V, the same result.

Meaning of K: at equilibrium $a(Zn^{2+}) / a(Cu^{2+}) \approx 10^{37}$. The reaction runs until practically no Cu^{2+} is left.

Such a huge K means the reaction goes practically to completion.

$\Delta G^\circ = -212.6$ kJ, $E^\circ_{cell} = +1.10$ V (d) $K \approx 1.8 \times 10^{37}$



4 · THE NERNST EQUATION

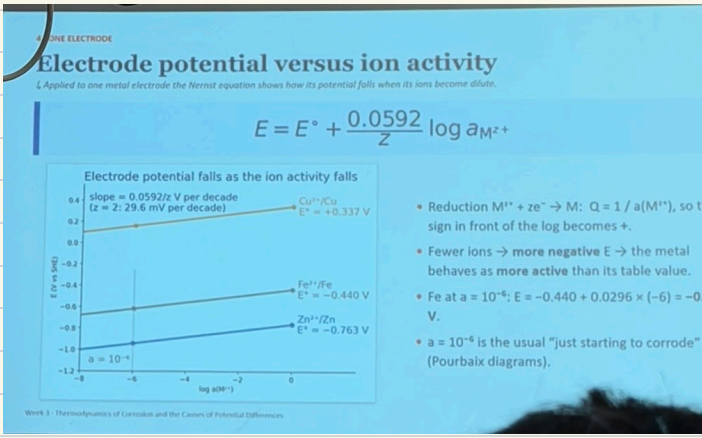
Potential at any concentration

Divide $\Delta G = \Delta G^\circ + RT \ln Q$ by −zF and you obtain the most useful equation of the course: the Nernst equation.

$$E = E^\circ - \frac{RT}{zF} \ln Q \qquad \text{at 25 °C:} \quad E = E^\circ - \frac{0.0592}{z} \log Q$$

- E: cell (or electrode) potential at the actual activities (V); E°: standard potential from the table (V); z: electrons reaction as written; Q: reaction quotient.
- R = 8.314 J/(mol·K), F = 96485 C/mol, T in kelvin. At 25 °C: 2.303·RT/F = **0.0592 V** (ln x = 2.303 log x).
- E° is the special case Q = 1 (log Q = 0).
- Use the 0.0592 form **only at 25 °C**; at another temperature use RT/zF with T in kelvin.

ln or log



ONE ELECTRODE

Electrode potential versus ion activity

Applied to one metal electrode the Nernst equation shows how its potential falls when its ions become dilute.

$$E = E^\circ + \frac{0.0592}{z} \log a_{M^{z+}}$$

Electrode potential falls as the ion activity falls

slope = 0.0592/z V per decade (z = 2: 29.6 mV per decade)

Cu^{2+}/Cu $E^\circ = +0.337$ V

Fe^{2+}/Fe $E^\circ = -0.440$ V

Zn^{2+}/Zn $E^\circ = -0.763$ V

a = 10^{-6}

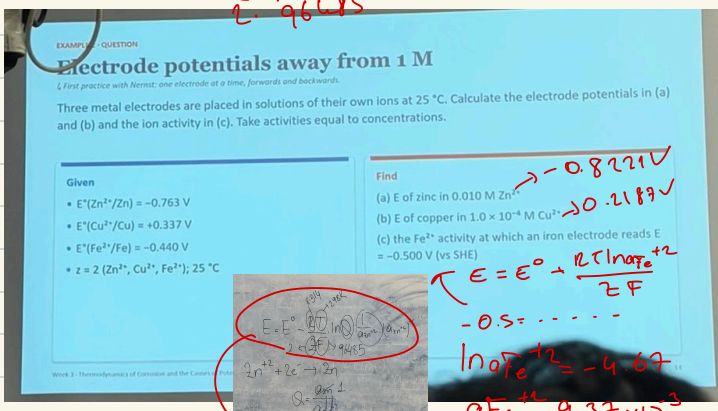
E (vs SHE) / V

log $a(M^{z+})$

- Reduction $M^{z+} + ze^- \rightarrow M$: Q = 1 / $a(M^{z+})$, so t sign in front of the log becomes +.
- Fewer ions → **more negative** E → the metal behaves as **more active** than its table value.
- Fe at a = 10^{-6}: E = −0.440 + 0.0296 × (−6) = −0 V.
- a = 10^{-6} is the usual "just starting to corrode" (Pourbaix diagrams).

Week 3 · Thermodynamics of Corrosion and the Causes of Potential Differences

$E = E^\circ + \frac{8.314}{2} \frac{298}{96485} \ln a_{Zn^{+2}}$



EXAMPLE 1 · QUESTION

Electrode potentials away from 1 M

First practice with Nernst: one electrode at a time, forwards and backwards.

Three metal electrodes are placed in solutions of their own ions at 25 °C. Calculate the electrode potentials in (a) and (b) and the ion activity in (c). Take activities equal to concentrations.

Given

- $E^\circ(Zn^{2+}/Zn) = -0.763$ V
- $E^\circ(Cu^{2+}/Cu) = +0.337$ V
- $E^\circ(Fe^{2+}/Fe) = -0.440$ V
- z = 2 (Zn^{2+}, Cu^{2+}, Fe^{2+}); 25 °C

Find

(a) E of zinc in 0.010 M Zn^{2+} → −0.8221 V

(b) E of copper in 1.0 × 10^{-4} M Cu^{2+} → 0.2189 V

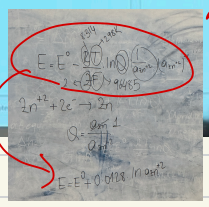
(c) the Fe^{2+} activity at which an iron electrode reads E = −0.500 V (vs SHE)

$E = E^\circ - \frac{RT \ln a_{Fe^{+2}}}{zF}$

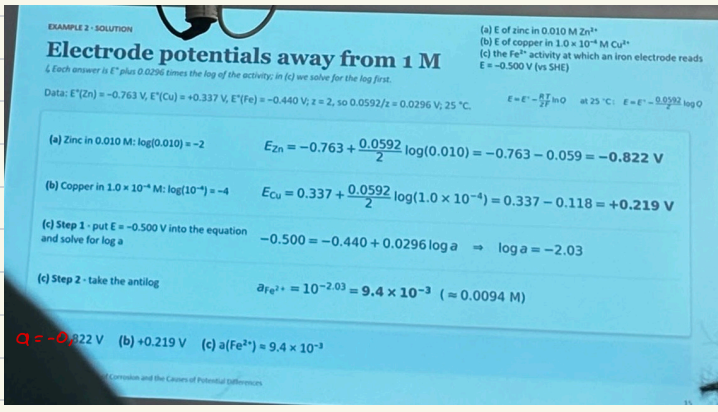
$-0.5 = \ldots$

$\ln a_{Fe^{+2}} = -4.67$

$a_{Fe^{+2}} = 9.37 \times 10^{-3}$



Week 3 · Thermodynamics of Corrosion and the Causes of Potential Differences



EXAMPLE 2 · SOLUTION

Electrode potentials away from 1 M

Each answer is E° plus 0.0296 times the log of the activity; in (c) we solve for the log first.

(a) E of zinc in 0.010 M Zn^{2+}
(b) E of copper in 1.0 × 10^{-4} M Cu^{2+}
(c) the Fe^{2+} activity at which an iron electrode reads E = −0.500 V (vs SHE)

Data: $E^\circ(Zn) = -0.763$ V, $E^\circ(Cu) = +0.337$ V, $E^\circ(Fe) = -0.440$ V; z = 2, so 0.0592/z = 0.0296 V; 25 °C.

$E = E^\circ - \frac{RT}{zF} \ln Q$ at 25 °C: $E = E^\circ - \frac{0.0592}{z} \log Q$

(a) Zinc in 0.010 M: log(0.010) = −2

$$E_{Zn} = -0.763 + \frac{0.0592}{2} \log(0.010) = -0.763 - 0.059 = -0.822 \text{ V}$$

(b) Copper in 1.0 × 10^{-4} M: log(10^{-4}) = −4

$$E_{Cu} = 0.337 + \frac{0.0592}{2} \log(1.0 \times 10^{-4}) = 0.337 - 0.118 = +0.219 \text{ V}$$

(c) Step 1 · put E = −0.500 V into the equation and solve for log a

$$-0.500 = -0.440 + 0.0296 \log a \quad \Rightarrow \quad \log a = -2.03$$

(c) Step 2 · take the antilog

$$a_{Fe^{2+}} = 10^{-2.03} = 9.4 \times 10^{-3} \quad (\approx 0.0094 \text{ M})$$

a = −0.822 V (b) +0.219 V (c) $a(Fe^{2+}) \approx 9.4 \times 10^{-3}$

Corrosion and the Causes of Potential Differences

Kimyasal potansiyel (μ)

Katı ve saf gazların "0" dır. (Fe, Cu, Zn, H_2, O_2)

$\Delta G^\circ = \Sigma N^\circ_{ürünler} - \Sigma N^\circ_{girenler}$

$\Delta G^\circ = -z \times F \times E^\circ$ $E>0$ $G<0$ spontane değil. $E<0$ $G>0$

$\Delta G = -RT \ln K$

$\Rightarrow -zFE^\circ = -RT \ln K \rightarrow E^\circ = \frac{RT \ln K}{zF}$

$\Rightarrow E^\circ = -\frac{\Delta G^\circ}{zF}$

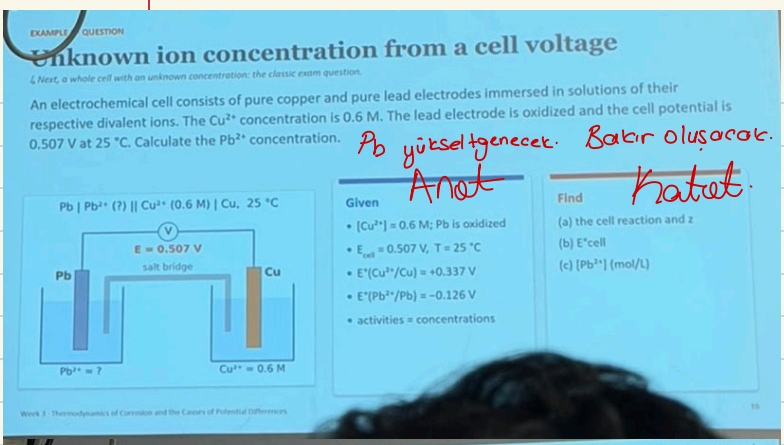
$\Delta G = \Delta G^\circ + RT \ln Q$

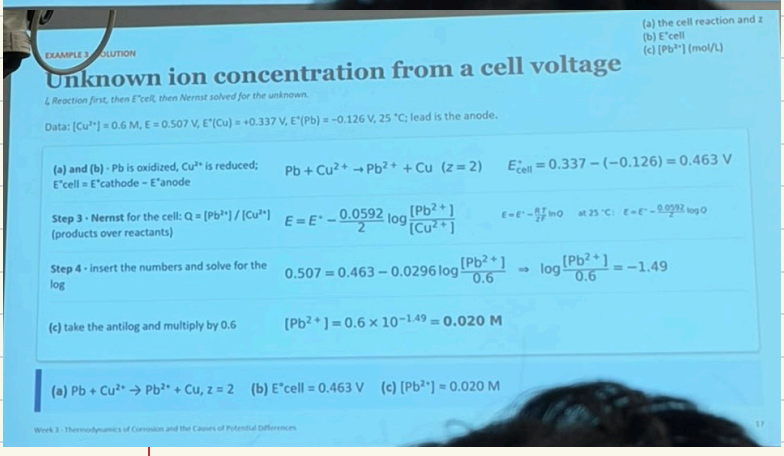
$-zFE = -zFE^\circ + RT \ln Q$ / "−zF"ye her taraf böl

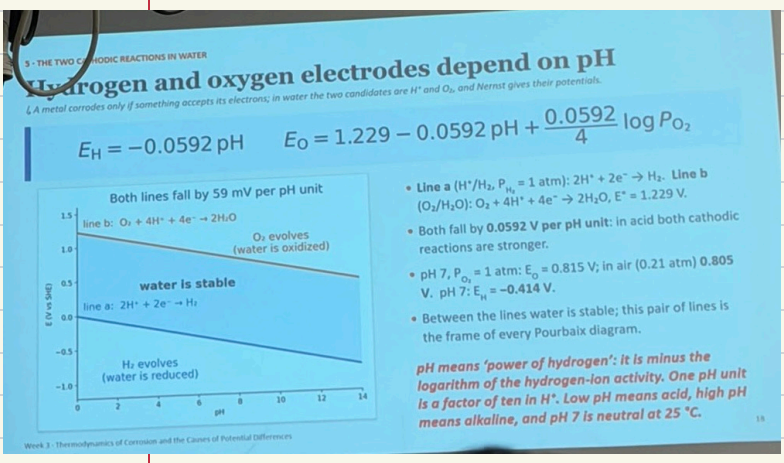
$E = E^\circ - \frac{RT \ln Q}{zF}$

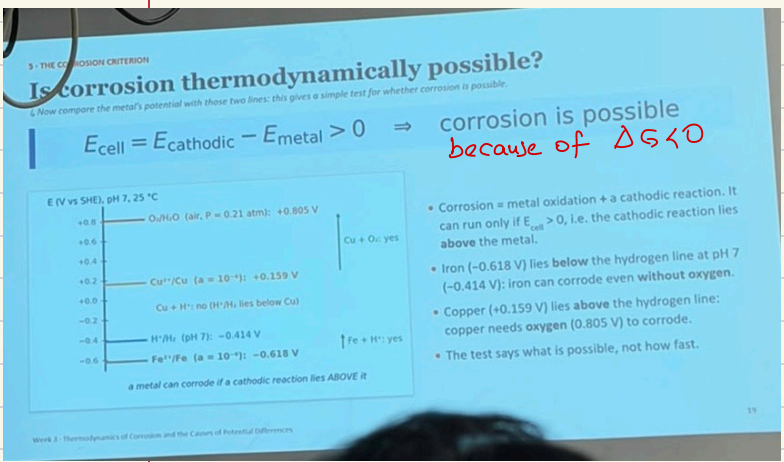
NERST DENKLEMİ

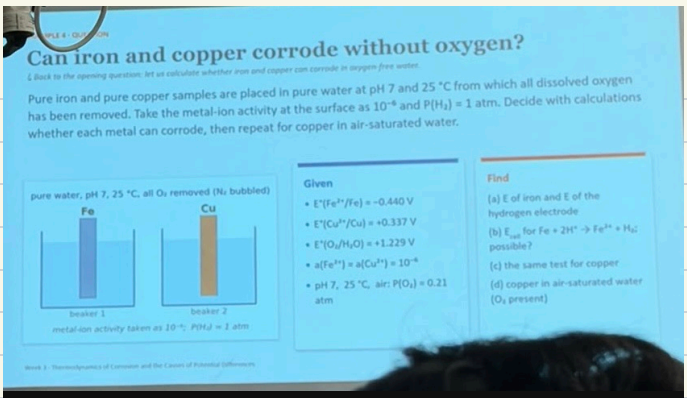
$Q = \frac{\text{ürünler}}{\text{girenler}}$











(a) Demirin ve Hidrojen Elektrodunun Gerçek Potansiyelleri (E)
Burada standart değerleri (E°) değil, Nernst denklemiyle o anki aktivitedeki (10^{-6}) gerçek değerleri buluyoruz. JPG

- **Demir (Fe) için:** $E_{Fe} = E^\circ + \frac{0.0592}{z}\log(a_{Fe^{2+}})$
  Demir 2 elektron verir ($z = 2$).
  $E_{Fe} = -0.440 + \frac{0.0592}{2}\log(10^{-6})$
  $E_{Fe} = -0.440 + 0.0296 \times (-6) = -0.440 - 0.1776 = \mathbf{-0.618}\ \text{V}$
- **Hidrojen (H_2) için (Oksijensiz suyun katodu):**
  Önceki slayttaki pratik formülü kullanıyoruz: $E_H = -0.0592 \times \text{pH}$
  $E_H = -0.0592 \times 7 = \mathbf{-0.414}\ \text{V}$

(b) Oksijensiz Ortamda Demir Korozyona Uğrar mı? ($Fe + 2H^+ \rightarrow Fe^{2+} + H_2$)
Korozyon olması için hücre potansiyelinin pozitif çıkması ($E_{cell} > 0$) gerekir.
Demir eridiği için anot, sudaki hidrojen gaz çıkardığı için katottur. JPG

- $E_{cell} = E_{katot} - E_{anot}$
- $E_{cell} = E_H - E_{Fe}$
- $E_{cell} = -0.414 - (-0.618) = \mathbf{+0.204}\ \text{V}$
  Sonuç: $E_{cell} > 0$ olduğu için reaksiyon spontanedir. **Evet, demir içinde hiç oksijen olmasa bile suda korozyona uğrar (paslanmasa da çözünür).**

(c) Aynı Testi Bakır İçin Yapalım
Önce bakırın o anki gerçek potansiyelini bulalım.

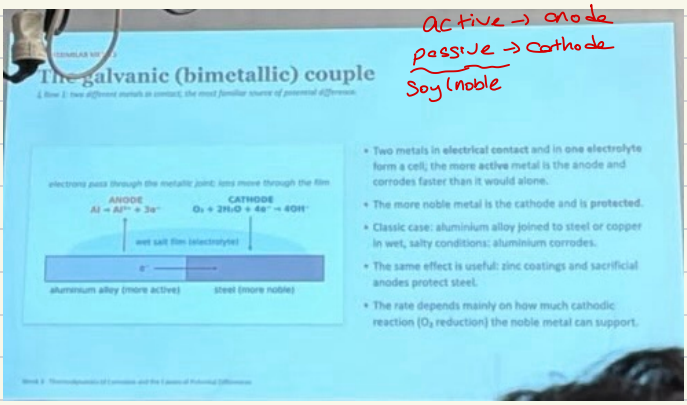
- **Bakır (Cu) için:** $E_{Cu} = E^\circ + \frac{0.0592}{2}\log(a_{Cu^{2+}})$
  $E_{Cu} = +0.337 + 0.0296 \times (-6) = +0.337 - 0.1776 = \mathbf{+0.159}\ \text{V}$

Şimdi hidrojene karşı hücre potansiyelini hesaplayalım:

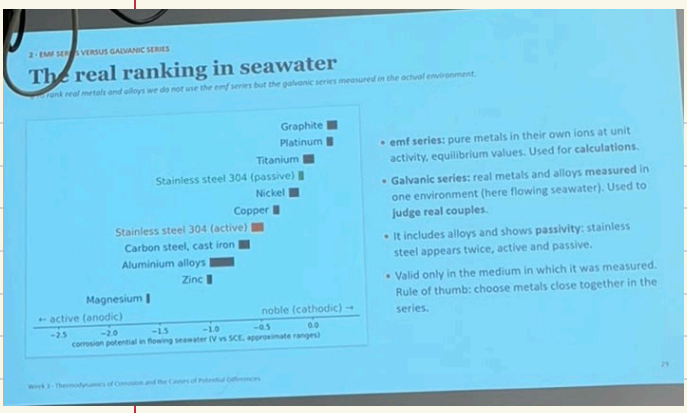
- $E_{cell} = E_H - E_{Cu}$
- $E_{cell} = -0.414 - (+0.159) = \mathbf{-0.573}\ \text{V}$
  Sonuç: $E_{cell} < 0$ çıktı. **Hayır, bakır oksijensiz saf suda kesinlikle korozyona uğramaz**

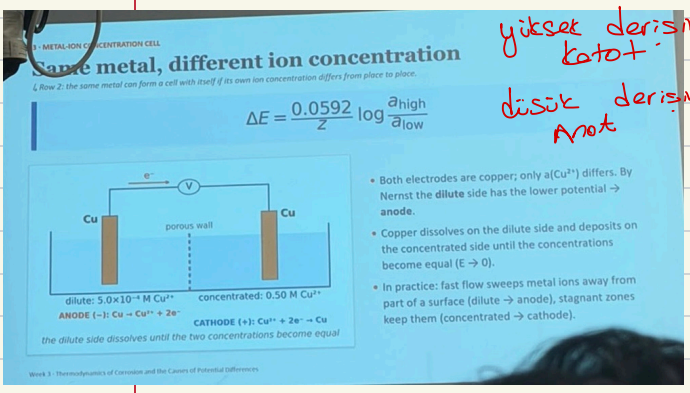
(d) Bakır Havalandırılmış (Oksijenli) Suda Korozyona Uğrar mı?
Ortamda oksijen ($P(O_2) = 0.21\ \text{atm}$) olduğunda katot artık hidrojen değil, oksijen reaksiyonu olur. JPG

- **Oksijenin Potansiyeli (E_O):** Önceki slayttaki uzun formülden hesaplarsak (pH 7 ve havada) yaklaşık $+0.805$ V çıkar. JPG
- Yeni Hücre Potansiyeli: $E_{cell} = E_{katot}(Oksijen) - E_{anot}(Bakır)$
- $E_{cell} = +0.805 - (+0.159) = \mathbf{+0.646}\ \text{V}$
  Sonuç: $E_{cell} > 0$ olduğu için reaksiyon artık spontanedir. **Evet, bakır ancak suya oksijen karıştığında korozyona uğrayabilir.**



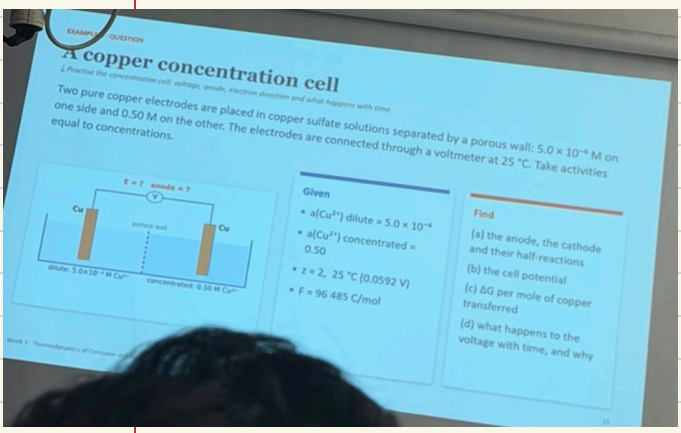
Galvaniz sadece element değil alaşımları da hesaplayabiliyor

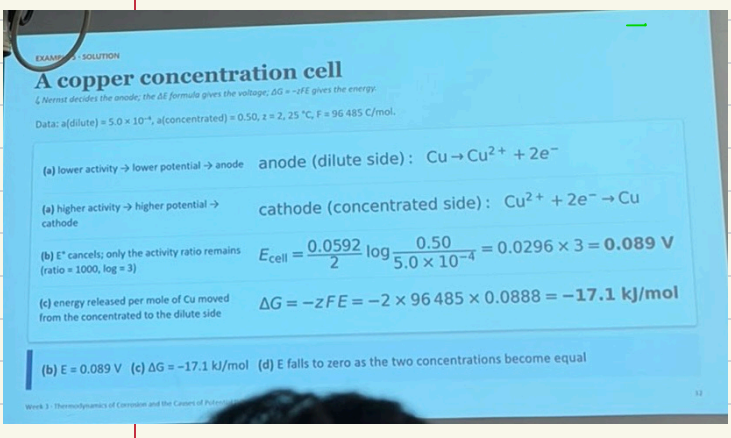




yüksek derişim katot.

düşük derişim Anot

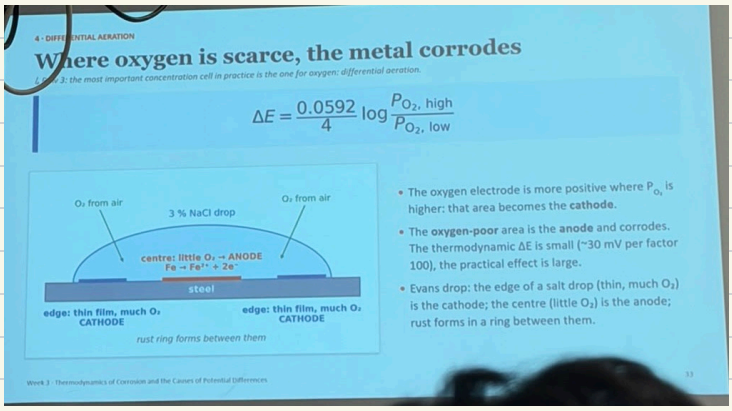


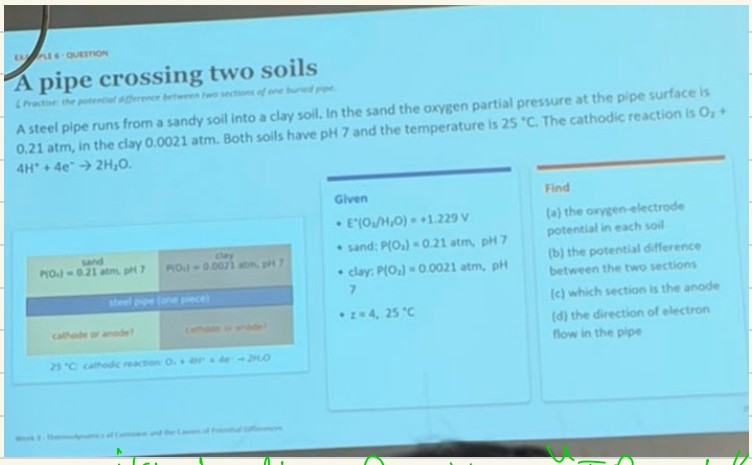


Derişik taraftaki iyonlar metal olarak birikir. Burada Cu^{+2} olarak harcanır. Yani reaksiyona girendir. Anotta Cu^{+2} oluşuyor.

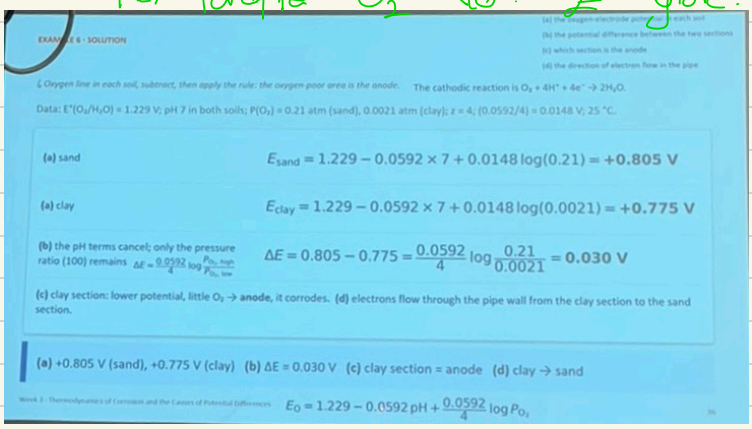
(Seyreltik / Derişik) buradan geliyor.

(Hepsini tam yazamadım çok hızlı söyledi.





İki tarafta O_2 var. E° yok!



→ İki tarafta da aynı metal varsa iyon derişimi daha düşük olan taraf anot olur. Çünkü onu yükseltgenip çözünmek ister.

→ İki taraf da Cu olduğu için E°'lar birbirini götürüyor. (Birbirine eşitler.)

$E_{cell} = E_{katot} - E_{anot}$ (25°C, 1 ATM, 8.314)

$E_{katot} = E^\circ_{katot} + \frac{0.0592}{z} \cdot \log(a_{derişik})$

$E_{anot} = E^\circ_{anot} + \frac{0.0592}{z} \cdot \log(a_{seyreltik})$

Çıkarma işlemi:

$E_{cell} = (E^\circ_{katot} - E^\circ_{anot}) + \frac{0.0592}{z} \log(a_{der.}) - \frac{0.0592}{z} \log(a_{sey.})$

($E^\circ_{katot} - E^\circ_{anot} = 0$)

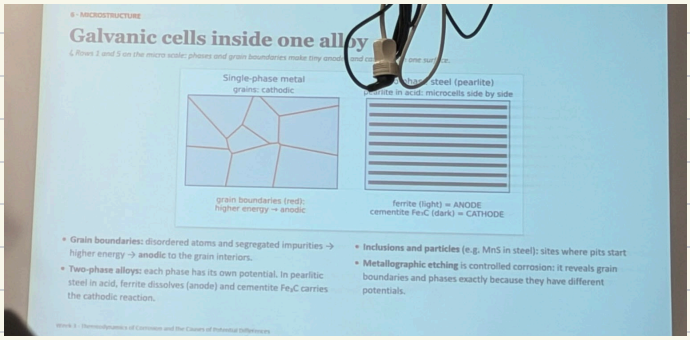
$E_{cell} = \frac{0.0592}{z} \cdot \log\left(\frac{a_{derişik}}{a_{seyreltik}}\right)$

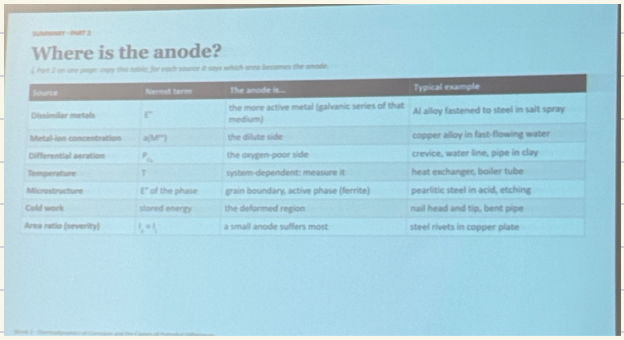
* tane sınırları daha aktif davranıyor.

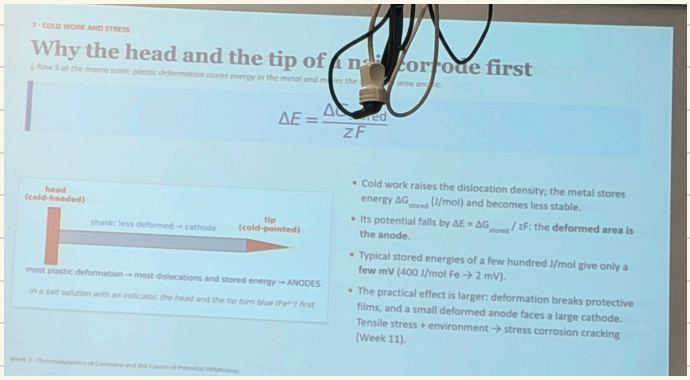
* tane sınırları daha aktif davranıyor.

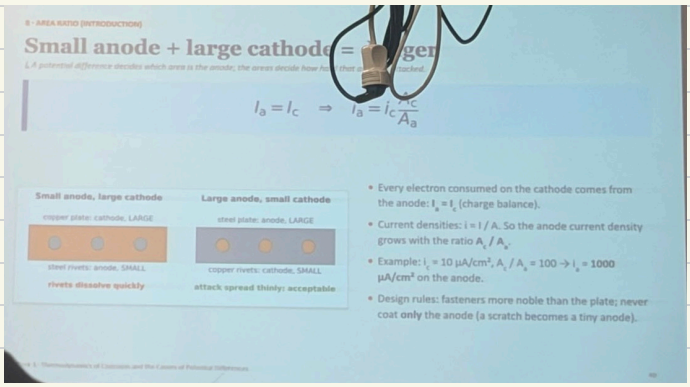
$$\Delta E = \frac{\Delta G_{stored}}{z \cdot F}$$

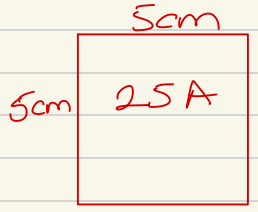
* korozyonun sebebi potansiyel farkın olması.



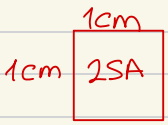








25 cm²
25 A



1cm²
2SA

$\frac{25\ A}{25\ cm^2} = 1\ A/cm^2$

$\frac{25\ A}{1\ cm^2} = 25\ A/cm^2$

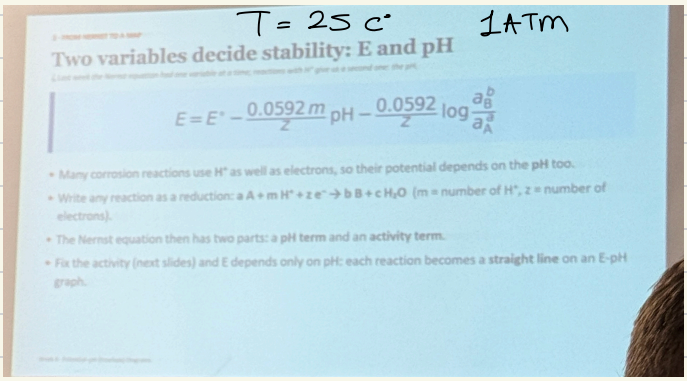
Why does a steel bar survive for decades inside wet concrete, but rust within weeks in a puddle of rain water? PH farkından.

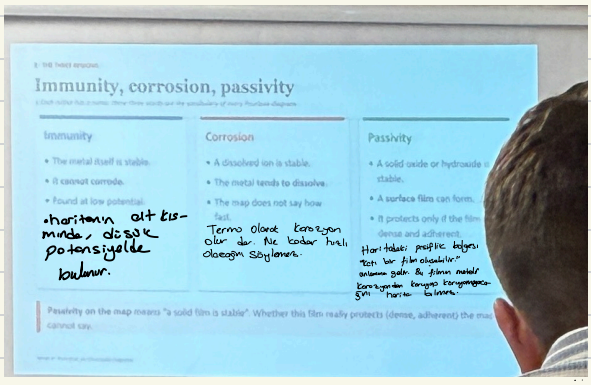
A map that tells which form of the metal is stable

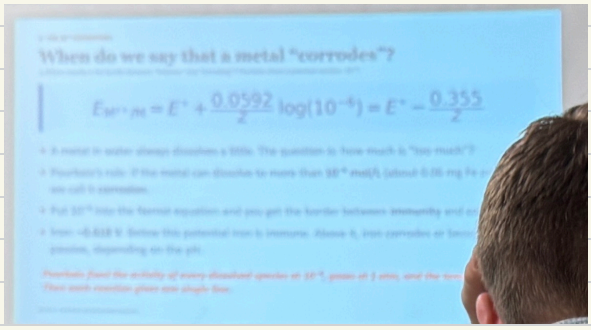
Origin
M. Pourbaix, PHD thesis, 1945.
later on atlas of diagrams for almost every element
Also called an E-pH diagram.

Like phase-Diagram
• Phase diagram: T and composition → the stable

Immunity: bağışıklılık. → reaksiyon yok
passivite: pasiflik → reaksiyon var ama yüzeyde koruyucu katman var.







MAVİ BÖLGE Immunity Area. Metal burada iyonlaşmak veya paslanmak istemez. Saf metal olarak kalır.

KIRMIZI BÖLGE: Corossion Area. Metalin e⁻ kaybettiği, korozyona uğradığı bölgedir. Sol → Asit / Sağ → Alkali

YEŞİL BÖLGE: Passivity Area. Orta kısım. Metal reaksiyona girer ama suda çözünmek yerine, yüzeyinde katı bir oksit veya hidroksit tabakası oluşturur. Bu tabaka metali korozyondan korur

ARA BÖLGE

Suyun alt sınırı

a çizgisinin altında su parçalanır. H_2 çıkar.
b " üstünde " " . O_2 çıkar.
Ara bölgede su kararlı.

1. Potansiyel (E) Aslında Ne Demektir?

- **Düşük/Eksi Potansiyel (Grafiğin Alt Tarafı):** Ortamda müthiş bir **elektron bolluğu ve itme gücü** vardır. Ortamdaki maddelere zorla elektron kabul ettirilmeye çalışılır (İndirgeyici ortam). PDF
- **Yüksek/Artı Potansiyel (Grafiğin Üst Tarafı):** Ortamda müthiş bir **elektron açlığı** vardır. Ortam, çevresindeki maddelerin elektronlarını sömürmek ve emmek ister (Yükseltgeyici ortam). PDF

2. Neden a Çizgisinin Altında Oksijen Değil de Hidrojen (H_2) Çıkar?

Grafikte aşağıya, yani **a çizgisinin altına** indiğinde ortamın potansiyelini aşırı düşürmüş olursun. Ortamda inanılmaz bir **elektron fazlalığı** oluşur. PDF

- Suda elektron kapmaya hazır ne var? Pozitif yüklü **H^+ iyonları**! PDF + 1
- H^+ iyonları bu fazla elektronları kapar ve nötrleşerek **Hidrojen gazına (H_2)** dönüşür:

$$2H^+ + 2e^- \rightarrow H_2$$

PDF + 1

- İşte bu yüzden a çizgisinin altında su parçalanır ve hidrojen kabarcıkları çıkar. a çizgisi hidrojene ait alt sınırdır. PDF + 3

3. Neden b Çizgisinin Üstünde Hidrojen Değil de Oksijen (O_2) Çıkar?

Grafikte yukarıya, yani **b çizgisinin üstüne** çıktığında potansiyel çok yükselir. Ortamda devasa bir **elektron açlığı** baş gösterir. PDF

- Ortam elektron emmek için sudaki maddeleri zorlar. Sudaki su molekülleri (H_2O) bünyesindeki oksijenin elektronlarını kaptırır.
- Su elektron vererek (yükseltgenerek) **Oksijen gazına (O_2)** dönüşür:

$$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$$

PDF + 1

- İşte bu yüzden b çizgisinin üstünde su oksitlenir ve oksijen gazı çıkar. b çizgisi de oksijene ait üst sınırdır. PDF + 3

4. Hidrojen Reaksiyonu Neden "Katodik"tir?

Elektrokimyadaki en temel iki tanım şudur: PDF + 1

- **Anot (Anodik):** Elektron **VEREN** (Yükseltgenen) taraftır. Metal çözünürken elektron fırlatır ($Fe \rightarrow Fe^{2+} + 2e^-$). PDF + 4
- **Katot (Katodik):** Elektron **ALAN / TÜKETEN** (İndirgenen) taraftır. PDF + 1

Bir metal korozyona uğrarken (anot olarak) elektron açığa çıkarır. Metalin korozyona devam edebilmesi için o serbest kalan elektronların **biri tarafından yutulması (alınması)** gerekir. PDF + 3

Suda gezen H^+ iyonları metalin fırlattığı bu elektronları **yuttuğu (aldığı)** için: PDF + 1

$$2H^+ + 2e^- \rightarrow H_2$$

Elektron tüketen taraf H^+ olduğu için bu reaksiyona **Katodik Reaksiyon** denir. PDF + 4

Ozetle Mantık Zinciri:

1. **Düşük voltaj (a'nın altı):** Elektron çok $\rightarrow$ H^+ elektron kapar $\rightarrow$ H_2 gazı çıkar **(Katodik İndirgenme)**. PDF + 1
2. **Yüksek voltaj (b'nin üstü):** Elektron açlığı var $\rightarrow$ Su elektronunu verir $\rightarrow$ O_2 gazı çıkar **(Anodik Yükseltgenme)**. PDF + 1
3. **İki çizginin arası:** Elektron açlığı da yok, bolluğu da yok $\rightarrow$ **Su sıvı (H_2O) olarak kararlıdır.**

1. b Çizgisi (Oksijen Çizgisi - Üst Sınır)

- **Yarı Reaksiyon:** $\mathrm{O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O}$ PDF + 1
- **Nernst Denklemi:** $E = 1.229 - 0.0592 \times \mathrm{pH}$ PDF + 1
- **Eğimi:** $-0.0592\ \mathrm{V/pH}$ (her 1 pH artışında potansiyel $59.2\ \mathrm{mV}$ düşer). PDF + 1
- **Kritik Sınır:**
  - $\mathrm{pH} = 0$ iken potansiyeli $+1.229\ \mathrm{V}$'tur. PDF + 1
  - $\mathrm{pH} = 14$ iken potansiyeli $+0.400\ \mathrm{V}$'a iner. PDF
  - b çizgisinin **üstündeki** potansiyellerde su molekülleri oksitlenir ve ortama **oksijen gazı ($\mathrm{O_2}$)** çıkar. PDF + 1

2. a Çizgisi (Hidrojen Çizgisi - Alt Sınır)

- **Yarı Reaksiyon:** $\mathrm{2H^+ + 2e^- \rightleftharpoons H_2}$ PDF + 1
- **Nernst Denklemi:** $E = 0.000 - 0.0592 \times \mathrm{pH} = -0.0592 \times \mathrm{pH}$ PDF + 1
- **Eğimi:** $-0.0592\ \mathrm{V/pH}$ (b çizgisine tamamen paraleldir). PDF + 1
- **Kritik Sınır:**
  - $\mathrm{pH} = 0$ iken potansiyeli $0.000\ \mathrm{V}$'tur. PDF + 1
  - $\mathrm{pH} = 14$ iken potansiyeli $-0.829\ \mathrm{V}$'a iner. PDF + 1
  - a çizgisinin **altındaki** potansiyellerde su indirgenir ve ortama **hidrojen gazı ($\mathrm{H_2}$)** çıkar.

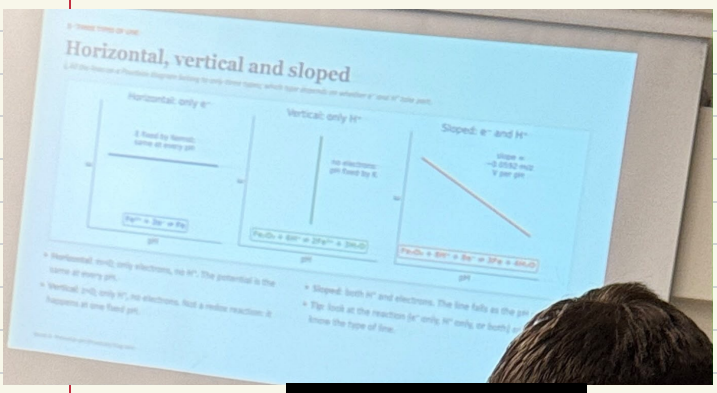
3. a ve b Çizgileri Arası (Suyun Kararlılık Bandı)

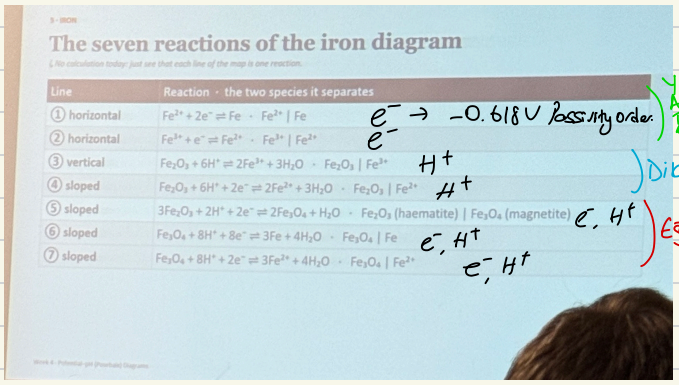
- **Suyun Kararlı Bölgesi:** a çizgisi ile b çizgisi arasında kalan açık mavi alan, suyun ayrışmadan sıvı $\mathrm{H_2O}$ şeklinde **kararlı olarak kaldığı** tek bölgedir. PDF + 1
- **Aralıktaki Sabit Voltaj Farkı ($\mathbf{1.23\ V}$):**
  Slaytın üzerinde kırmızı kalemle çizilen ok, iki çizgi arasındaki dikey potansiyel farkını ($1.229\ \mathrm{V} - 0.000\ \mathrm{V} \approx 1.23\ \mathrm{V}$) gösterir. İki çizgi birbirine tamamen paralel olduğu için her pH değerinde aralarındaki potansiyel farkı daima $1.23\ \mathrm{V}$'tur. PDF + 3

4. Bazik (Alkali) Ortamda Reaksiyon Yazımı

- Grafiğin sağ tarafında ($\mathrm{pH} > 7$, yani alkali ortamlar) reaksiyonlar $\mathrm{H^+}$ yerine $\mathrm{OH^-}$ iyonları cinsinden yazılır: HTML
  - Katodik (Hidrojen): $\mathrm{2H_2O + 2e^- \rightleftharpoons H_2 + 2OH^-}$ HTML
  - Anodik (Oksijen): $\mathrm{O_2 + 2H_2O + 4e^- \rightleftharpoons 4OH^-}$ HTML
- Denklemdeki iyon türü değişse de çizgilerin konumu, eğimi ve değerleri birebir aynı kalır.

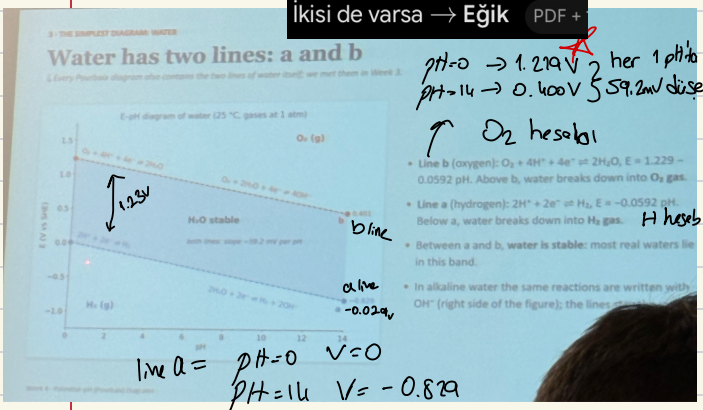
horizontal → yatay
vertical → dikey

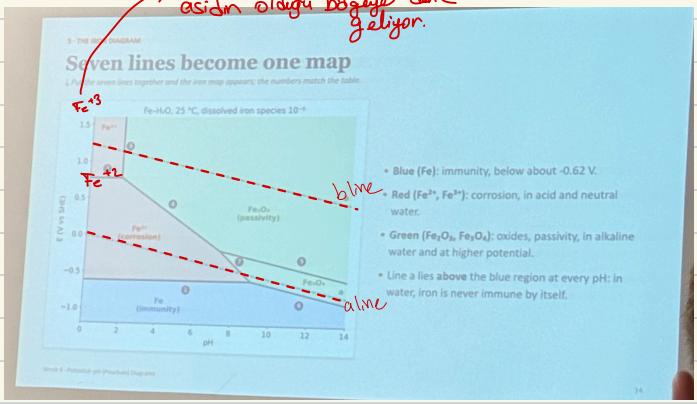


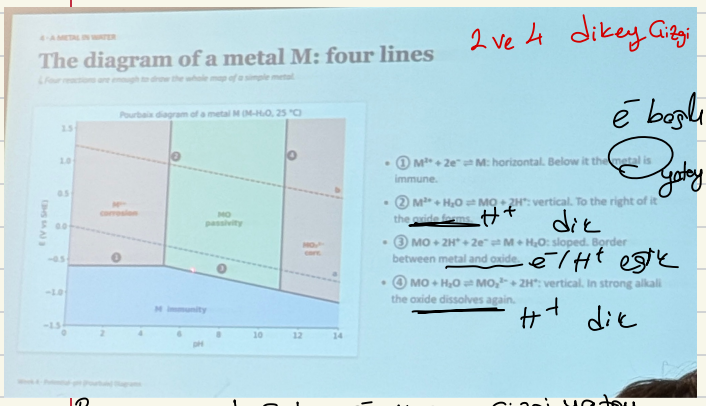


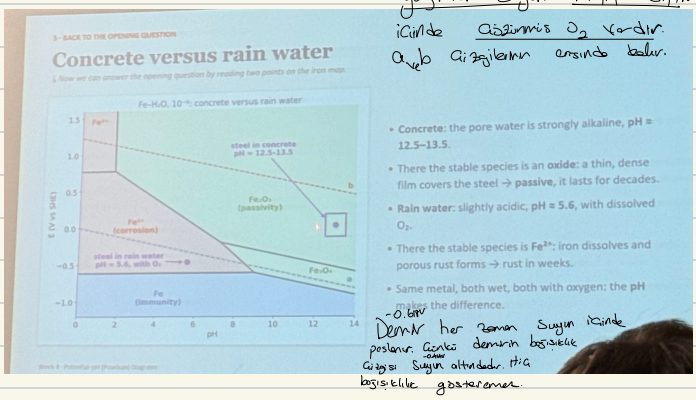
Sadece e^- varsa → Yatay
Sadece H^+ varsa → Dikey
İkisi de varsa → Eğik PDF +

IRON





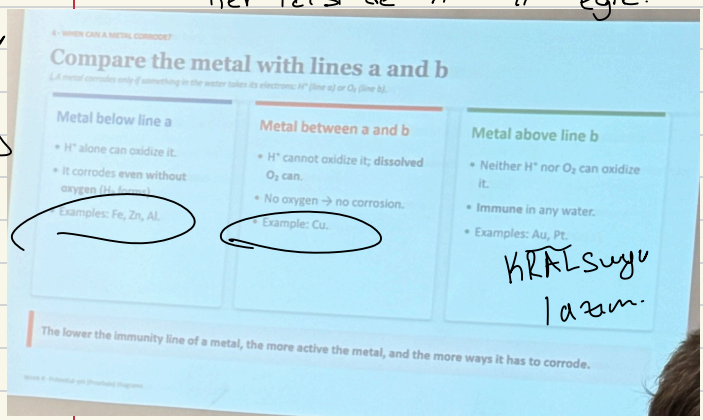


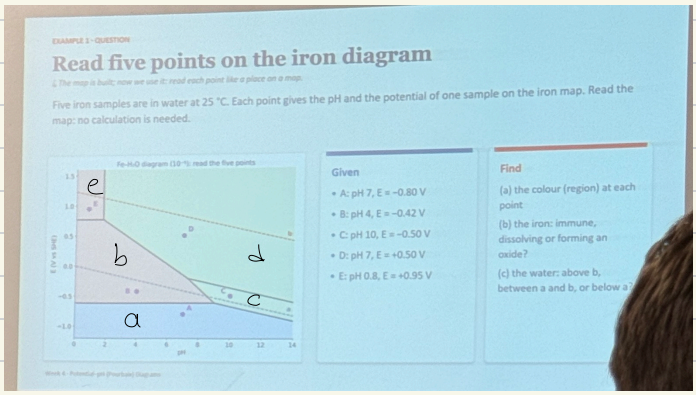


yağmur suyun hafif asitte içinde çözünmüş O_2 vardır. a ve b çizgilerinin arasında kalır.

Demir her zaman suyun içinde paslanır. Çünkü demirin bozulmuş çizgisi suyun altındadır. Bu bozulmuşluk gösterilmemiş.

Bir reaksiyonda Sadece e^- varsa çizgi yatay
" H^+ " " dikey
her ikisi de " " eğik.





Batırın korozyonu için O_2'ye ihtiyaç duyarım.
Demir için O_2 olursa korozyon olur.
Olmazsa da yavaş olur ama olur.

a) Immunity region → Pure IRON,

b) IRON divalent ions , corossion region

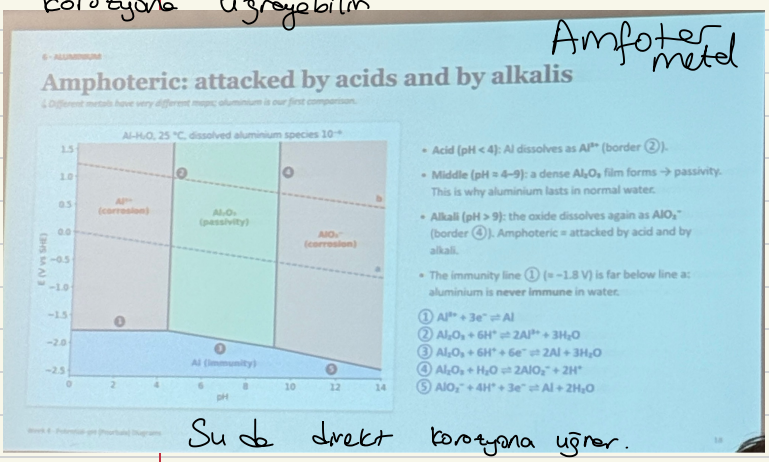
c) Passivity Region , Magnetite

d) Passivity Region, Hematite

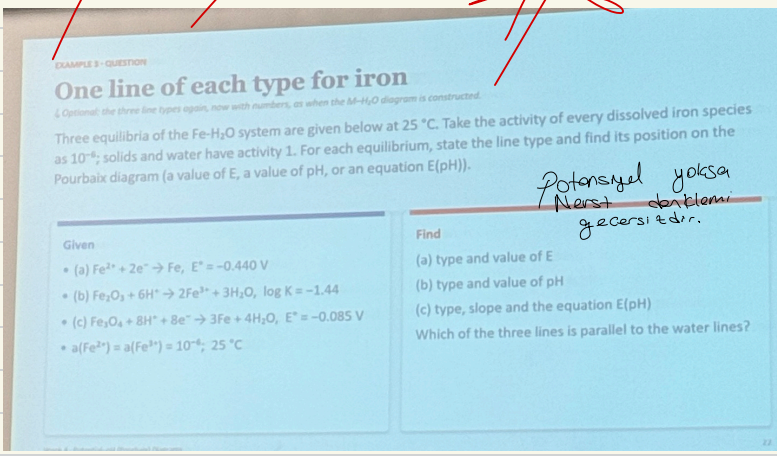
e) Corrosion Region, IRON Trivalent Ions

Alumme

hem asit, hem de bazlar tarafından korozyona uğrayabilir

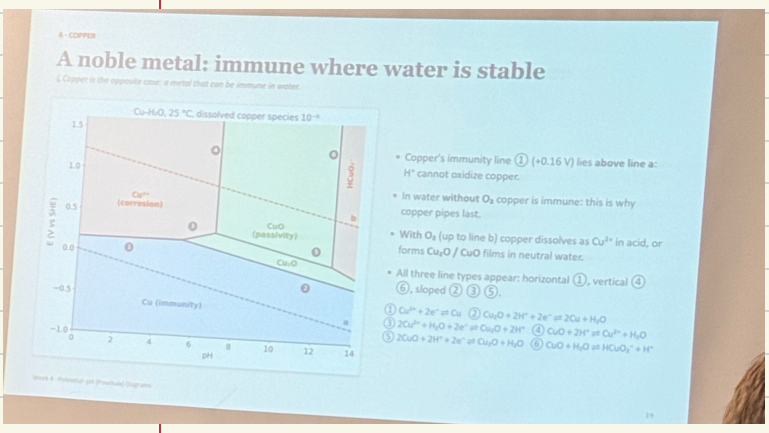


Su da direkt korozyona uğrar.

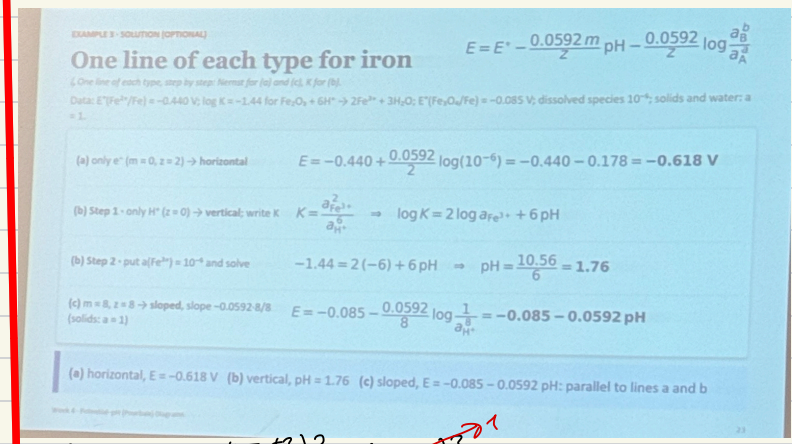


Potansiyel yoksa Nernst denklemi geçersizdir.

Copper



*"H" bakırı oksitleyemez. Oksitlenmesi için "O_2" ihtiyaç var.



b) $K = \frac{a(Fe^{+3})^2 \cdot a(H_2O)^3}{a(Fe_2O_3) \cdot a(H^+)^6} \Rightarrow K = \frac{a(Fe^{+3})^2}{a(H^+)^6}$

her tarafın log'unu al.

kimyada $-\log a(H^+) = pH$ demek.

$\log K = \log a(Fe^{+3})^2 - \log a(H^+)$

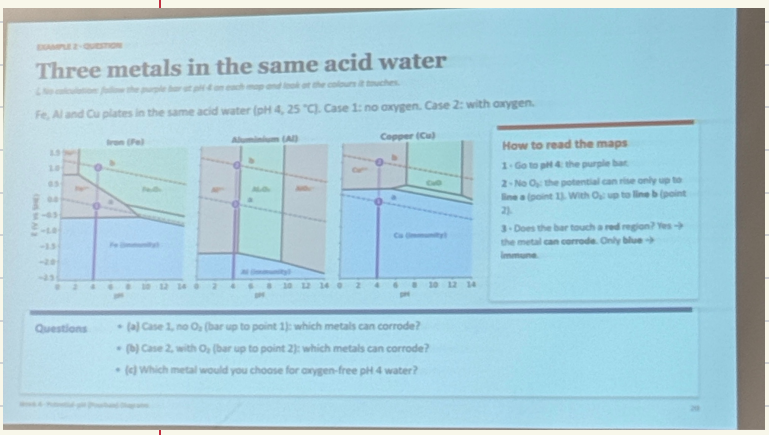
$= 2\log a_{Fe^{+3}} - 6\log a(H^+)$

$6\,pH$

$\log K = 2\log a_{Fe^{3+}} + 6pH$

$-1.44 = 2(-6) + 6pH$ $\quad 6pH = 10.56$

$pH = 1.76$



a) $E = E^\circ - \frac{0.0592}{2}\log\left(\frac{a_{Fe}}{a_{Fe^{2+}}}\right)$ $\quad a_{Fe} = 1$, $a_{Fe^{2+}} = 10^{-6}$

$= -0.440 - \frac{0.0592}{2}\cdot\log\left(\frac{1}{10^{-6}}\right)$

$\log 10^6 = 6\log 10 = 6$

$= -0.440 - \frac{0.0592 \cdot 6}{2}$

$= -0.440 - 0.178 = -0.618$ V

a) IRON, Aluminum → "O_2" make Corrosion Copper cant. → Bakır kullanırım

B) Hepsi korozyona uğrar.

C) I will choose Copper.

Bu soru için

anlamadığım şu . a ve b şıklarında formülleri seçtik mi seçtiysek nasıl seçtik bu nerst denklemenin bir sürü çeşidi var ben anlamakta zorlanıyorum hangisini kullanacağımın. mesela cevap kısmının sağ üstünde bi formül yazmış o formülü neden kullanmadık

Thought for 18s ›

Çok haklısın; elektrokimyada Nernst denkleminin bazen artılı (+), bazen eksili (−), bazen logaritmanın içi ters dönmüş halleri kullanılır ve bu durum kafa karıştırabilir.

Aslında **cevap kısmının sağ üstünde yazan o formülü KULLANDIK!**

Farklı farklı formüller seçmedik; o sağ üstteki formül **"Ana Şablon"**dur. Sadece bazı tepkimelerde o formülün bazı parçaları **sıfır olup yok olduğu için** sanki bambaşka formüller kullanılmış gibi görünür.

Gel, o sağ üstteki formülü parçalarına ayıralım ve a ile b şıkkında ne olduğunu tek tek görelim.

Sağ Üstteki "Ana Formül" Neyi Anlatıyor?

Sağ üstteki formül şudur:

$$E = E^\circ - \underbrace{\frac{0.0592 \cdot m}{z}\text{pH}}_{\text{pH (yani H}^+\text{) Parçası}} - \underbrace{\frac{0.0592}{z}\log\frac{a_B^b}{a_A^a}}_{\text{Diğer İyonların Parçası}}$$

Buradaki harflerin anlamı:

- z: Tepkimeye giren/çıkan **elektron (e^-) sayısı**
- m: Tepkimeye giren/çıkan **proton (H^+) sayısı**
- a_B/a_A: H^+ dışındaki ürün ve giren iyonların derişimleri/aktiviteleri.

1) (a) Şıkkında Sağ Üstteki Formülü Nasıl Kullandık?

Tepkimemiz: $\text{Fe}^{2+} + 2e^- \rightarrow \text{Fe}$

Şimdi ana formüldeki değişkenlere bakalım:

- Elektron sayısı: $z = 2$
- Proton (H^+) sayısı: $m = 0$ (Tepkimede hiç H^+ yok!)
- İyonlar: Ürünlerde katı demir var ($a_B = 1$), girenlerde Fe^{2+} var ($a_A = 10^{-6}$).

Hadi bunları sağ üstteki formüle koyalım:

$$E = -0.440 - \left(\frac{0.0592 \times \mathbf{0}}{2} \times \text{pH}\right) - \frac{0.0592}{2}\log\left(\frac{1}{10^{-6}}\right)$$

Fark ettiysen ortadaki pH **kısmı 0 ile çarpıldığı için tamamen yok oldu!**

Geriye sadece şu kaldı:

$$E = -0.440 - \frac{0.0592}{2}\log\left(\frac{1}{10^{-6}}\right)$$

> **Buradaki artı/eksi kafanı karıştırmasın:** Matematikten biliyoruz ki $-\log(1/x) = +\log(x)$'tir. Hoca çözüm kağıdında $-\log\left(\frac{1}{10^{-6}}\right)$ yazmak yerine doğrudan işaretleri ters çevirip $+\log(10^{-6})$ yazmış:
>
> $$E = -0.440 + \frac{0.0592}{2}\log(10^{-6})$$
>
> İkisi matematiksel olarak **birebir aynı şeydir**.

Gördüğün gibi (a) şıkkında farklı bir formül seçmedik, sağ üstteki formülde $m = 0$ yazdığımız için formül kendiliğinden kısaldı.

2) (b) Şıkkında Neden Sağ Üstteki Formülü KULLANAMADIK?

Tepkimemiz: $\text{Fe}_2\text{O}_3 + 6\text{H}^+ \rightarrow 2\text{Fe}^{3+} + 3\text{H}_2\text{O}$

Şimdi ana formüle koymayı deneyelim:

- Bu tepkimede kaç tane elektron (e^-) var? **HİÇ YOK ($z = 0$)!**
- Sağ üstteki formüle bak: z sayısı **paydada** yer alıyor ($\frac{0.0592}{z}$).
- Paydaya 0 yazarsan ne olur? $\frac{0.0592}{0} \rightarrow$ **Tanımsız olur!** Matematiksel olarak formül çöker.
- Ayrıca soruda bu tepkime için E° (potansiyel) verilmiş mi? **Hayır, $\log K$ verilmiş.**

İşte kağıdın sağ üstüne yazılan **"Potansiyel yoksa Nernst denklemi geçersizdir"** notu tam olarak bunu anlatır.

Elektron transferi yoksa bu bir redoks (elektrokimya) olayı değildir; sıradan bir kimyasal dengedir. Bu yüzden Nernst formülü yerine, kimyadaki klasik denge bağıntısı olan $K = \frac{[\text{Ürünler}]}{[\text{Girenler}]}$ formülü kullanılır:

$$K = \frac{[\text{Fe}^{3+}]^2}{[\text{H}^+]^6}$$

Logaritmasını aldığında da çözümdeki ifadeye ulaşırsın.

3) (c) Şıkkında Ne Oldu?

Tepkimemiz: $\text{Fe}_3\text{O}_4 + 8\text{H}^+ + 8e^- \rightarrow 3\text{Fe} + 4\text{H}_2\text{O}$

- Elektron var mı? Evet, $z = 8$.
- Proton (H^+) var mı? Evet, $m = 8$.
- Katılar ve suyun aktivitesi 1 olduğu için $\log\left(\frac{a_B}{a_A}\right) = \log(1) = 0$ olur.
- Bu sefer sağ üstteki formülün **en sağdaki iyon kısmı sıfır olup gitti**, ortadaki pH kısmı kaldı:

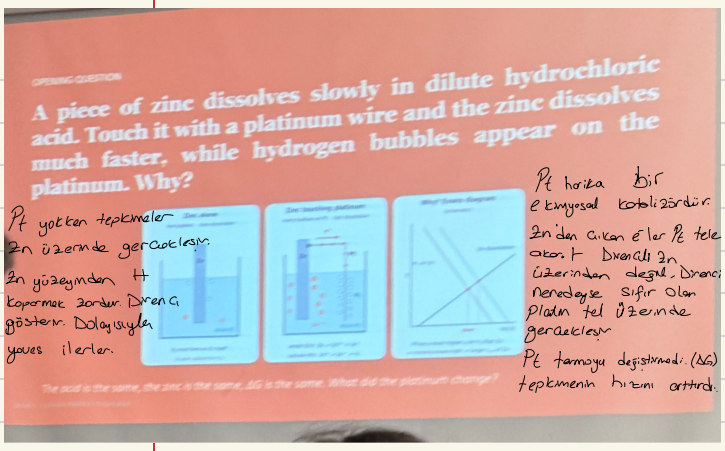
$$E = E^\circ - \frac{0.0592 \times 8}{8}\text{pH} = -0.085 - 0.0592\,\text{pH}$$

📌 Sınav İçin Pratik Karar Kuralı

Bir soru gördüğünde hangi yolu seçeceğini anlamak için kendine sadece şu 2 soruyu sor:

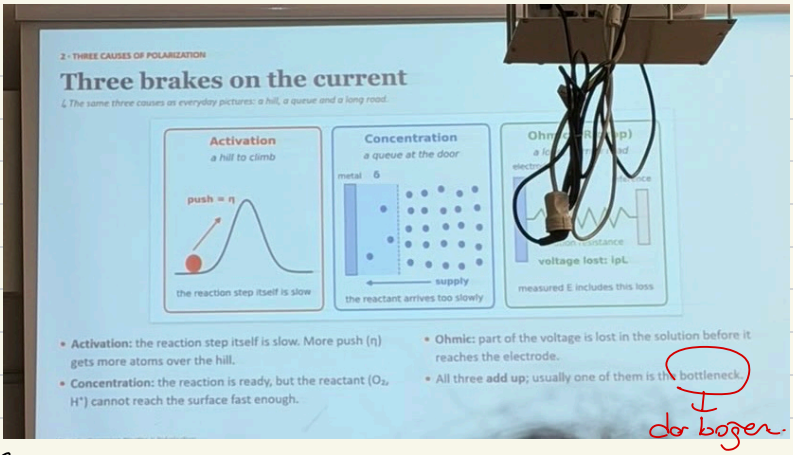
Tepkimede Ne Var?	Formül	Açıklama
Sadece e^- var, H^+ yok (Şık a)	Sağ üstteki formülü al, pH terimini sil ($m = 0$).	Sonuç: Sadece sayı çıkar ($E =$ sabit).
Sadece H^+ var, e^- yok (Şık b)	Nernst'i çöpe at! Denge formülü yaz ($K = \frac{\text{Ürün}}{\text{Giren}}$).	Sonuç: Sadece pH çıkar ($\text{pH} =$ sabit).
Hem e^- hem H^+ var (Şık c)	Sağ üstteki formülü aynen kullan.	Sonuç: $E(\text{pH})$ denklemi çıkar.

Anodik (yükseltgenme): $Zn \rightarrow Zn^{2+} + 2e^-$ (Gmlco çözünür)
katotik (indirgenme): $2H^+ + 2e^- \rightarrow H_2\uparrow$ (H gazı olarak çıkar)

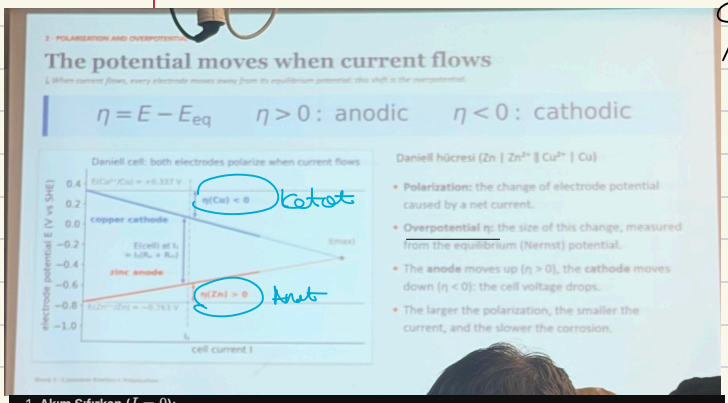


Pt yokken tepkimeler Zn üzerinde gerçekleşir.
Zn yüzeyinden H koparmak zordur. Direnç gösterir. Dolayısıyla yavaş ilerler.

Pt hariç bir elektrokimyasal katalizördür.
Zn'den çıkan e'ler Pt tele akar. Direnci Zn üzerinden değil, direnci neredeyse sıfır olan Platin tel üzerinde gerçekleşir.
Pt tamamen değişmedi. (ΔG) tepkimenin hızını arttırdı.



dar boğaz.



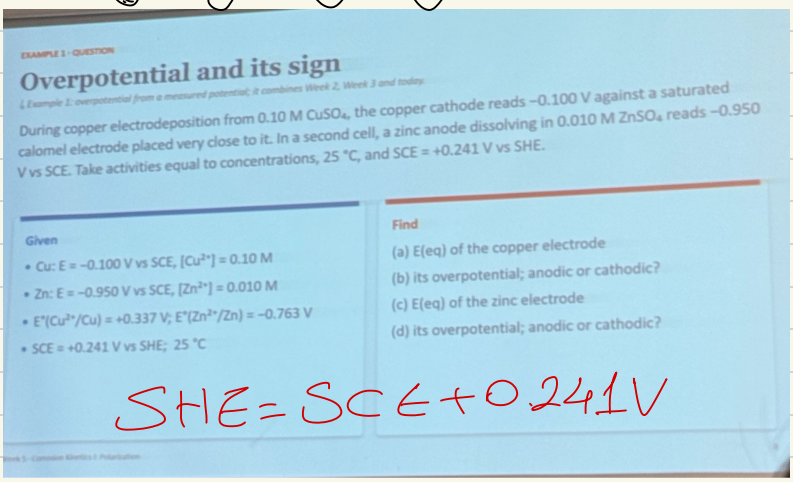
katot

anot

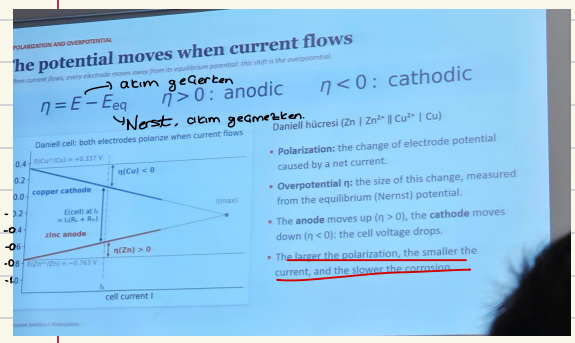
1. **Akım Sıfırken ($I = 0$):**
   - Bakır katot dengededir: $E(Cu^{2+}/Cu) = +0.337$ V
   - Çinko anot dengededir: $E(Zn^{2+}/Zn) = -0.763$ V
   - İki uç arasındaki açık devre voltajı yaklaşık 1.1 V'tur.
2. **Akım Akmaya Başladığında ($I > 0$):**
   - **Anot yukarı doğru kayar ($\eta > 0$):** Çinkonun potansiyeli -0.763 V'tan daha pozitif değerlere doğru tırmanır.
   - **Katot aşağı doğru kayar ($\eta < 0$):** Bakırın potansiyeli $+0.337$ V'tan daha negatif değerlere doğru düşer.
   - **Voltaj Çöküşü:** İki elektrot potansiyeli birbirine doğru yaklaşır, yani pilin ürettiği net voltaj ($E_{katot} - E_{anot}$) akım arttıkça azalır.
3. **Maksimum Akım (I_{max} - Kısa Devre):**
   - Direnç sıfır olduğunda iki çizgi kesişir. Burası çekilebilecek en büyük akımdır (korozyonda bu, serbest korozyon akımı I_{corr} olur).

$\eta = 0$ olacak denge sağlanacak.

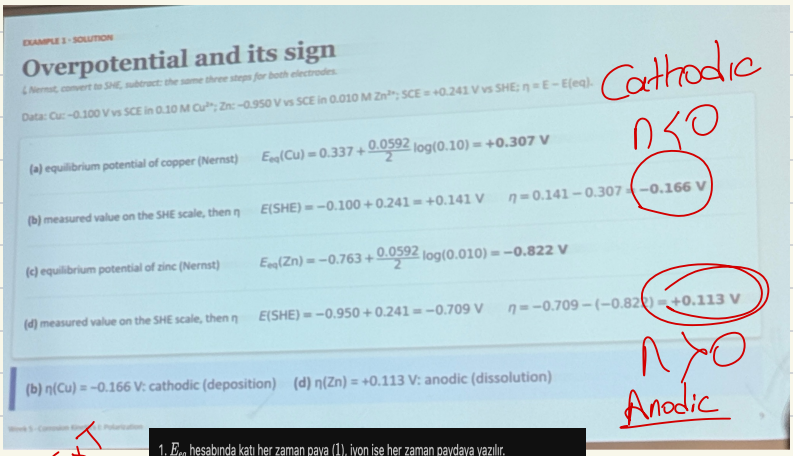
Soru 2 · Aşırı gerilim ve işareti (derste Örnek 1). 0.10 M $CuSO_4$'ten bakır kaplama sırasında bakır katot, ona çok yakın yerleştirilmiş doymuş kalomel elektroda karşı −0.100 V okunuyor. İkinci bir hücrede, 0.010 M $ZnSO_4$ içinde çözünen çinko anot SCE'ye karşı −0.950 V okunuyor. Aktiviteleri konsantrasyona eşit alın. Bulun: (a) bakır elektrodun E(eq) değeri, (b) aşırı gerilimi: anodik mi katodik mi?, (c) çinko elektrodun E(eq) değeri, (d) aşırı gerilimi: anodik mi katodik mi?



SHE = SCE + 0.241 V



akım geçerken Nernst, akım geçmezken.



Cathodic η<0

η>0 Anodic

① $\Delta G = \Delta G^\circ + RT\ln Q$

② $\Delta G = -zFE$

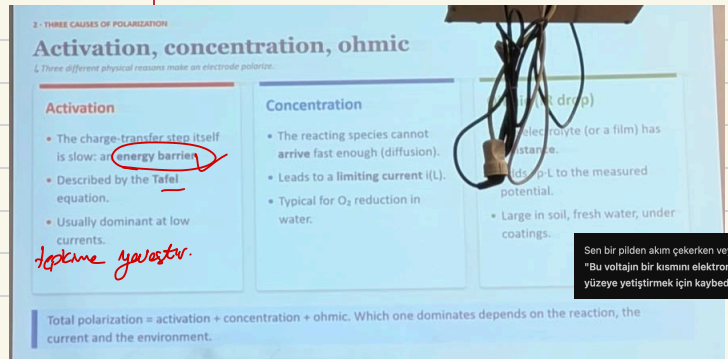
$E = E^\circ - \frac{RT}{zF}\cdot \ln Q$

$E = E^\circ - \frac{2.303 \times R \times T}{zF}\cdot \log Q$

Polarizasyon FAZLAYSA (çizgiler DİK):
→ Hemen toslarlar → **Akım AZ** → Korozyon AZ (metal korunur).
Polarizasyon AZ / DÜŞÜKSE (çizgiler YATAY):
(Cümlende "poli azsa" demek istedin)
→ X ekseninde taaa sağa doğru giderler → **Akım ÇOK FAZLA** → Metal fena erir

1. E_{eq} hesabında katı her zaman paya (1), iyon ise her zaman paydaya yazılır.
2. SCE gördüğün anda hemen +0.241 ekleyip SHE yapmadan asla η çıkarma işlemine geçme!
3. η **eksi** çıkarsa katot (kaplanma), **artı** çıkarsa anot (erime/çözünme) demektir.

$m = \frac{M \cdot I_{akim} \cdot t}{x \cdot F}$ ↑ Faraday Nernst kısmı çıkar.



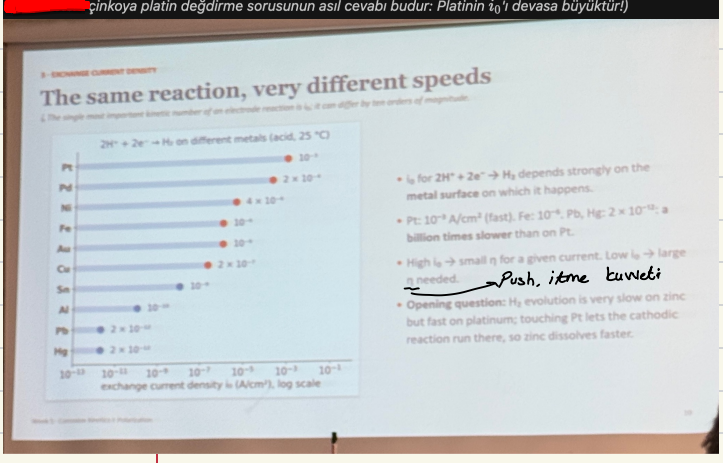
tepkime yavaştır.

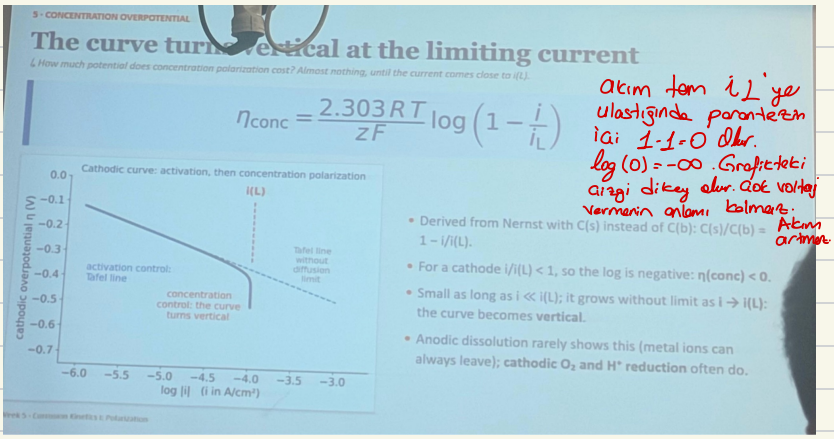
Over potential, polarization Ohmic reply? Nedir gibi sorular çıkar.

Sen bir pilden akım çekerken veya bir metal paslanırken voltaj kaybedersin. Bu slayt diyor ki:
"Bu voltajın bir kısmını elektronları zıplatmaya zorlamak için kaybedersin (Aktivasyon), bir kısmını hammaddeyi yüzeye yetiştirmek için kaybedersin (Konsantrasyon), bir kısmını da sıvının iç direncine kaptırırsın (Ohmik)."

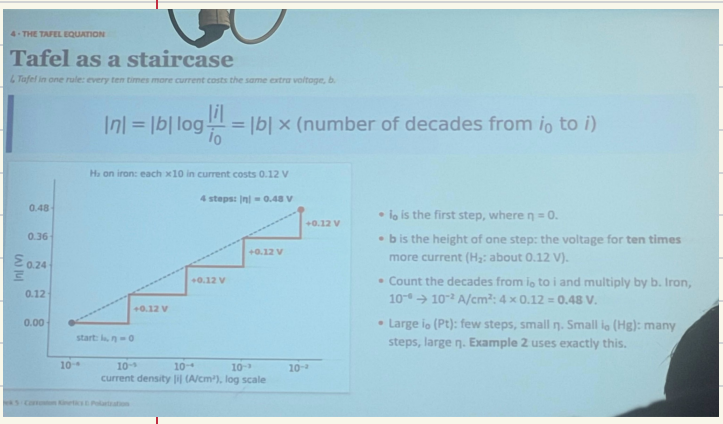
Tek Cümlelik Kural:
i_0 **ne kadar BÜYÜKSE**, reaksiyon o kadar kolay yürür → Arkadan itmek için gereken voltaj cezası (η) o kadar **KÜCÜK** olur.

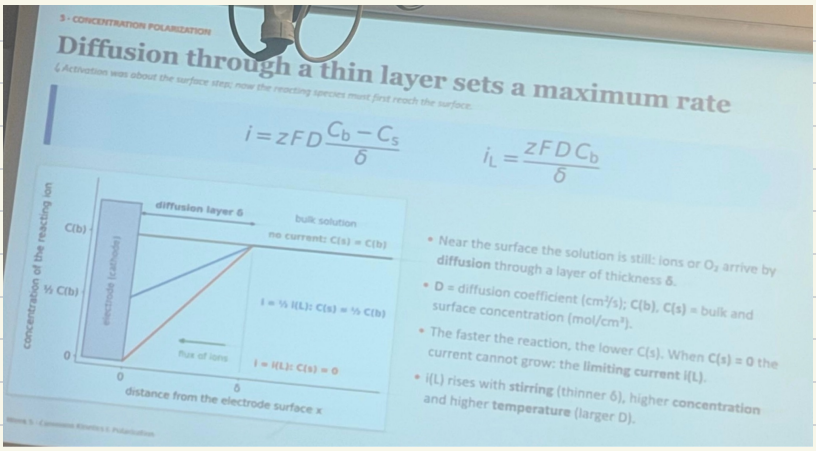
[redacted]çinkoya platin değdirme sorusunun asıl cevabı budur: Platinin i_0'ı devasa büyüktür!)



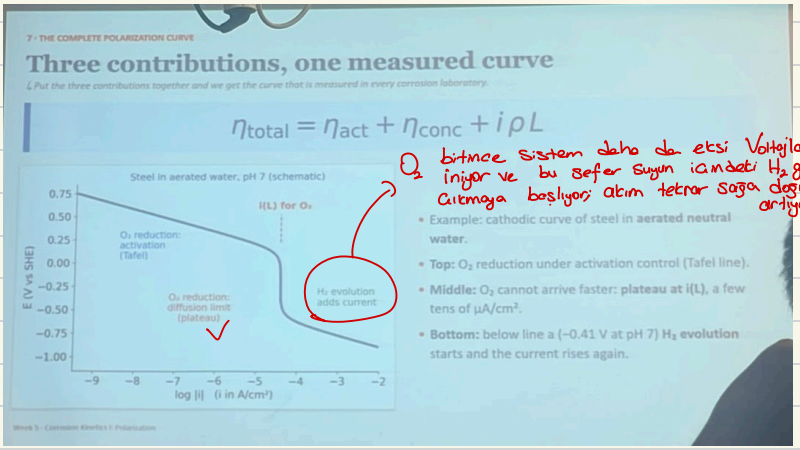


Akım tam i_L'ye ulaştığında paranteren içi 1-1=0 olur. log(0) = −∞. Grafikteki dikey olur. Çok voltaj vermenin anlamı kalmaz. Akım artmaz.



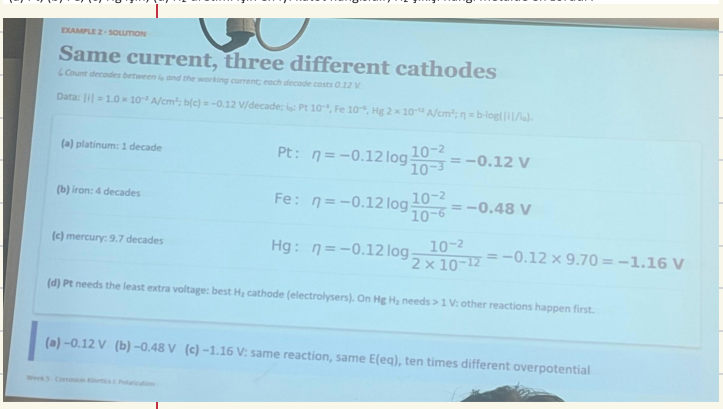


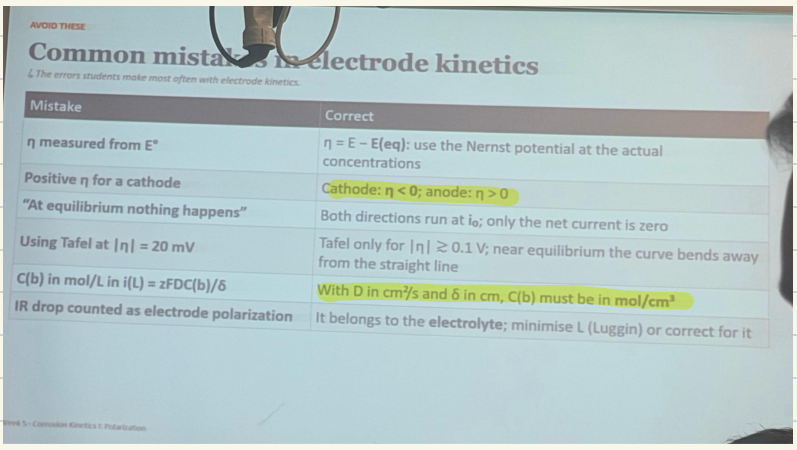
1. i_0: Tepkimenin başlangıç çizgisidir. Ne kadar büyükse o kadar iyidir (daha az voltaj kaybedersin).
2. **Tafel** (b): Akımı 10 kat artırmak için ödediğin basamak parasıdır (0.12 V). Kaç basamak çıkarsan o kadar b ödersin.

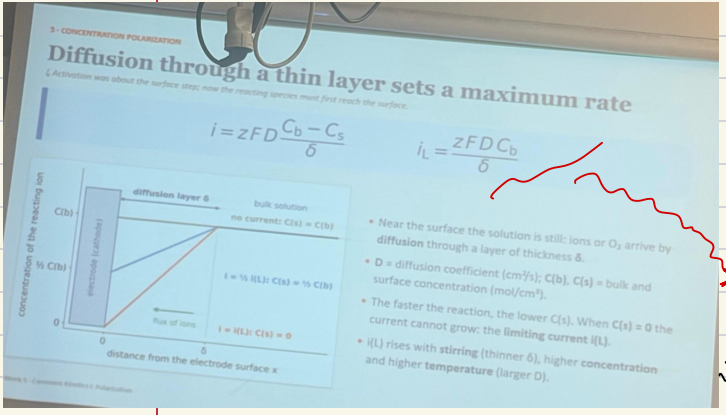


O₂ bitince sistem daha da eksi voltajlara iniyor ve bu sefer suyun içindeki H_2 gazı çıkmaya başlıyor, akım tekrar sağa doğru artıyor.

Soru 3 · Aynı akım, üç farklı katot (derste Örnek 2). Asit içinde platin, demir ve cıva üzerinde 10 mA/cm² (1.0×10^{-2} A/cm²) katodik akım yoğunluğuyla hidrojen çıkıyor. Üçü için b(c) = −0.12 V/dekat, i_0 değerlerini tablodan alın. Bulun: η, (a) Pt, (b) Fe, (c) Hg için; (d) H_2 üretimi için en iyi katot hangisidir, H_2 çıkışı hangi metalde en zordur?



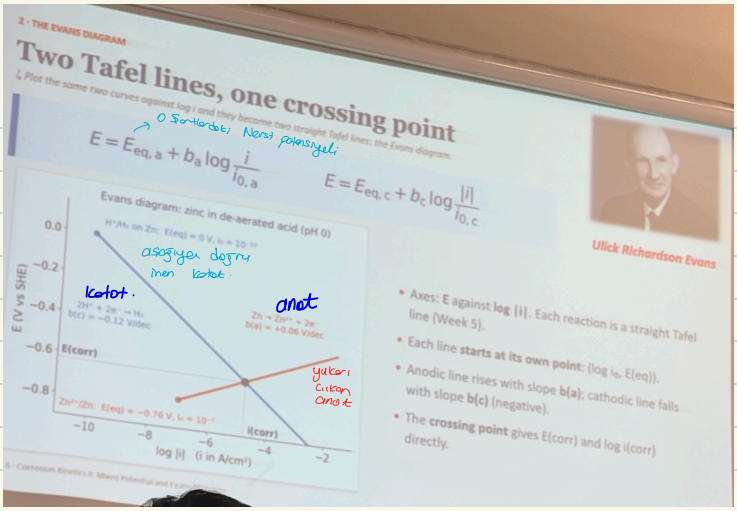




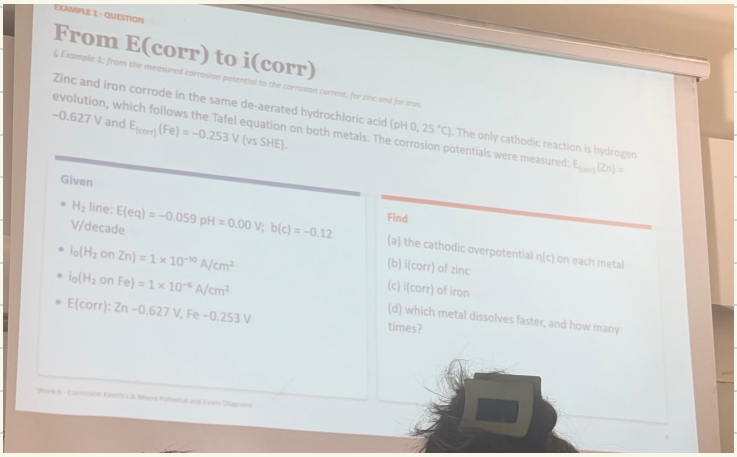
δ↓ i_L↑

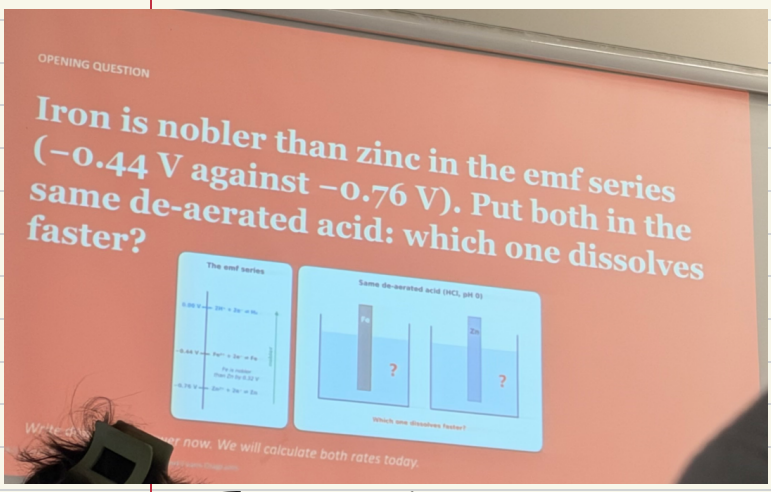
Çözeltiyi karıştırırsan tabaka incelir (δ küçülür) ve bu limit akım tavanı yukarıya çıkar.

* Corossion olması için anodik reaksiyonun karşılığında bir katodik reaksiyonunun olması gerekir.



Soru:

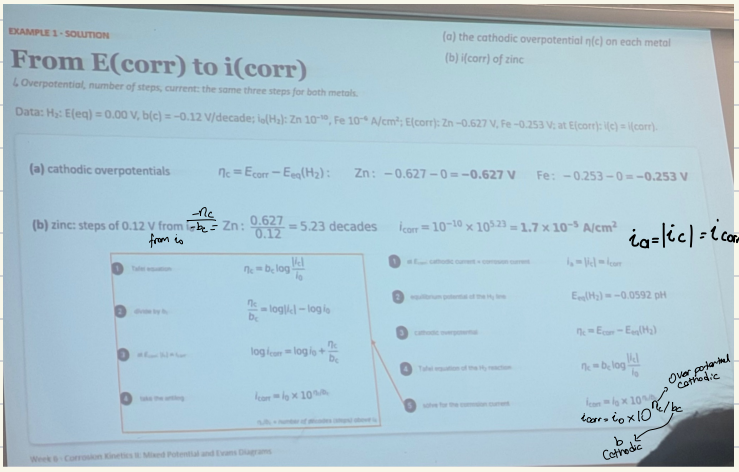




$Fe \rightarrow Fe^{+2} + 2e^-$ (anodic)

$H^{+2} + 2e^- \longrightarrow H_2$ (cathodic)

Ne kadar Anodic reaksiyon
O kadar Cathodic reaksiyon.

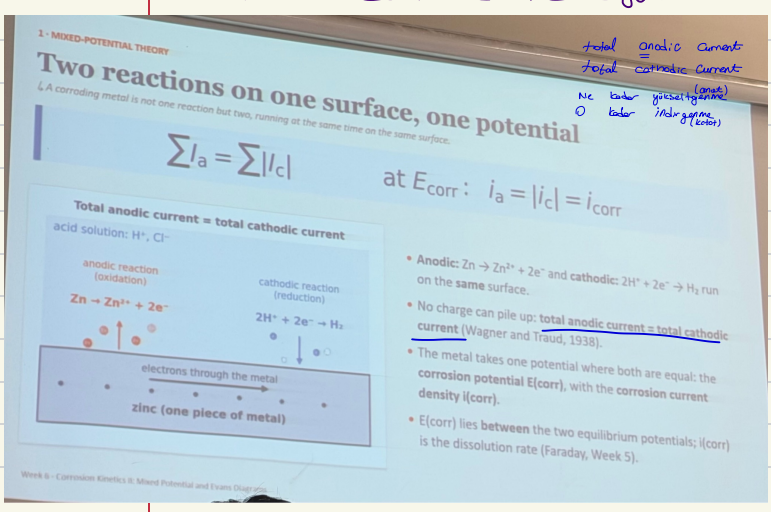


C)

$$\frac{-\eta_c}{-b_c} = \frac{-0.253}{+0.12} = 2.10 \text{ decades}$$

$$i_{corr} = i_0 \times 10^{2.10}$$

$$= 1.2 \times 10^{-4} \text{ A/cm}^2$$



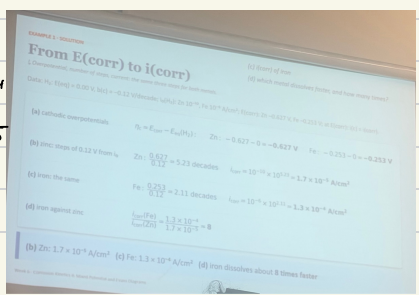
total anodic current
total cathodic current
Ne kadar yüksetgenler (anot)
O kadar indirgenme (katot)

D)

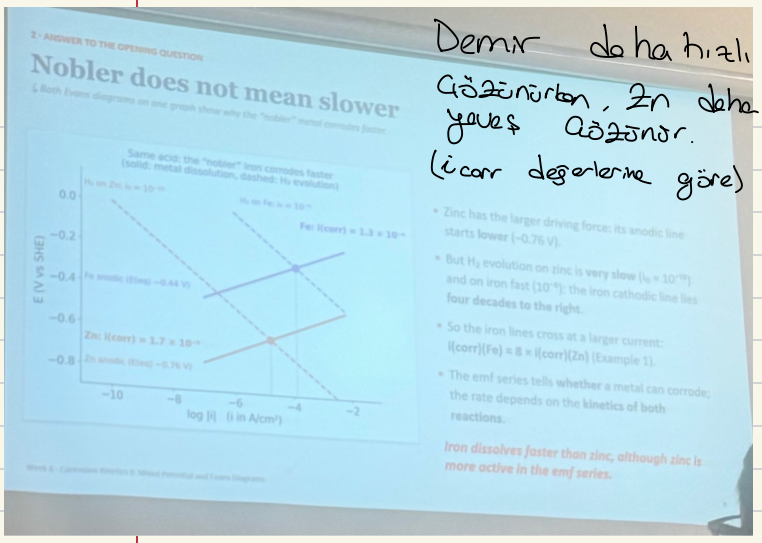
$$\frac{i_{corr}(Fe)}{i_{corr}(Zn)} = \frac{1.3 \times 10^{-4}}{1.7 \times 10^{-5}}$$

$$= 8$$

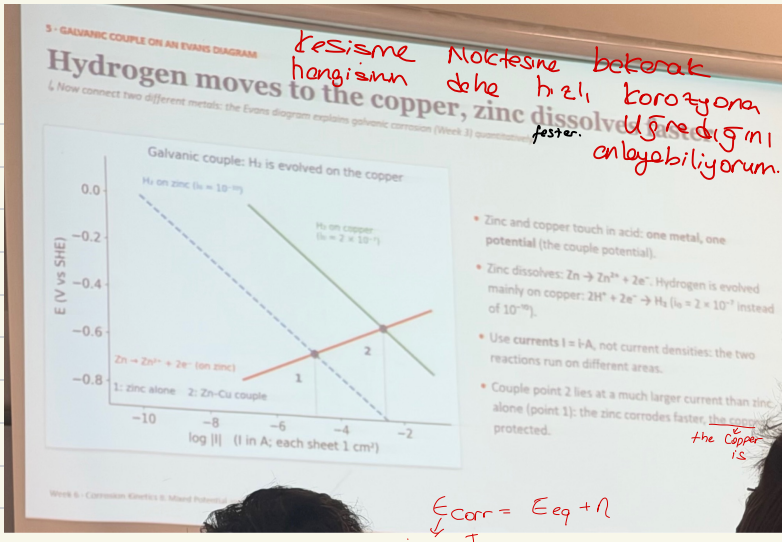
iron dissolves about 8 times faster.



IEVAMI



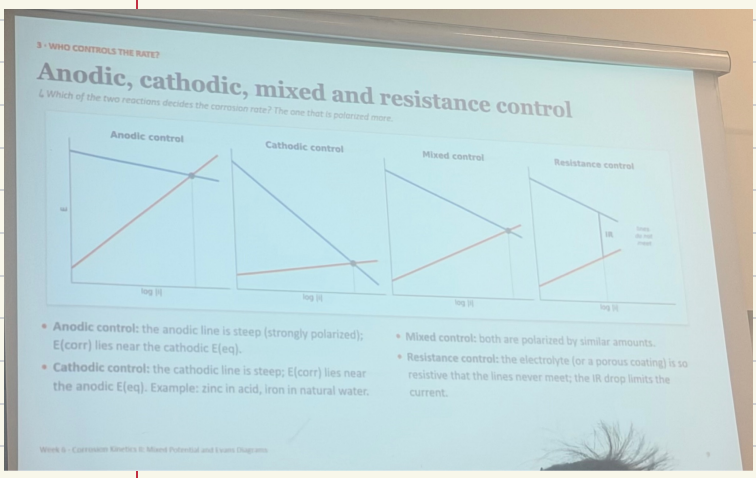
Demir daha hızlı çözünürken, Zn daha yavaş çözünür. (i corr değerlerine göre)

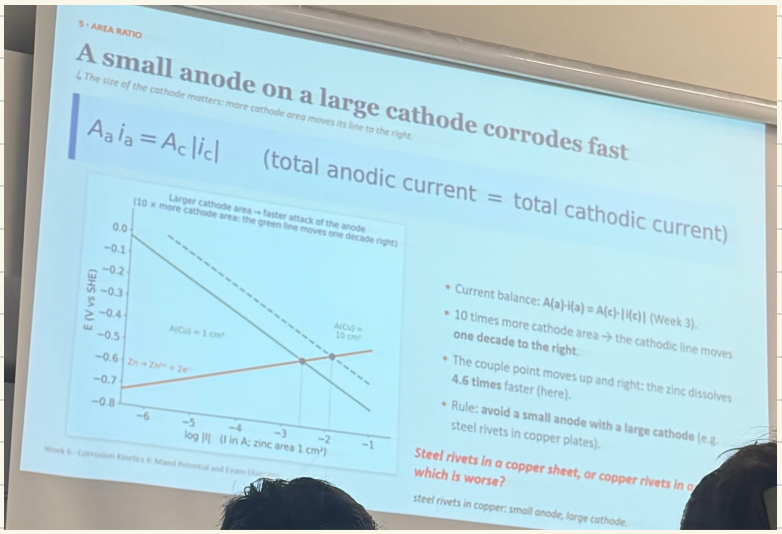


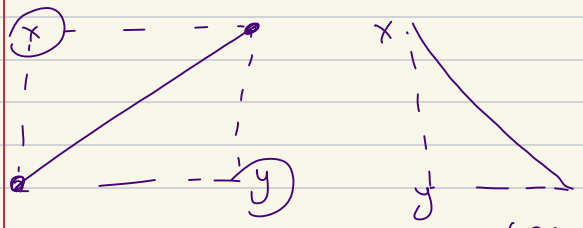
Direnç Noktasına bakarak daha hızlı korozyona uğradığını anlayabiliyorum.

$E_{corr} = E_{eq} + \eta$

$i_0 \rightarrow \frac{I}{A}$







x ve y değerleri

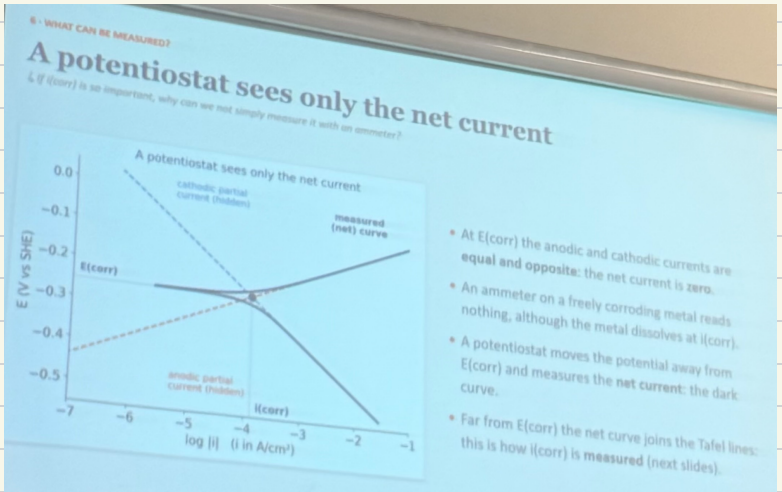
daha dik olan eğri bu reaksiyonun daha kolay olduğunu kontrol eder.

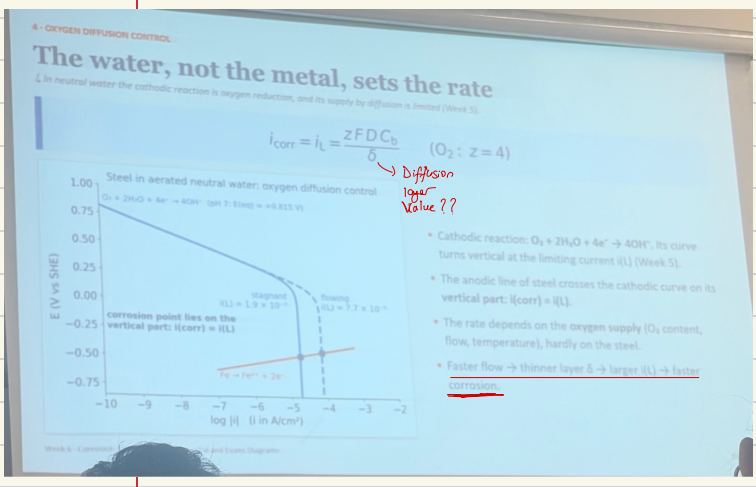
x-y yaptım bu bana over potential (η) değerini verdi. eğer bu değeri;

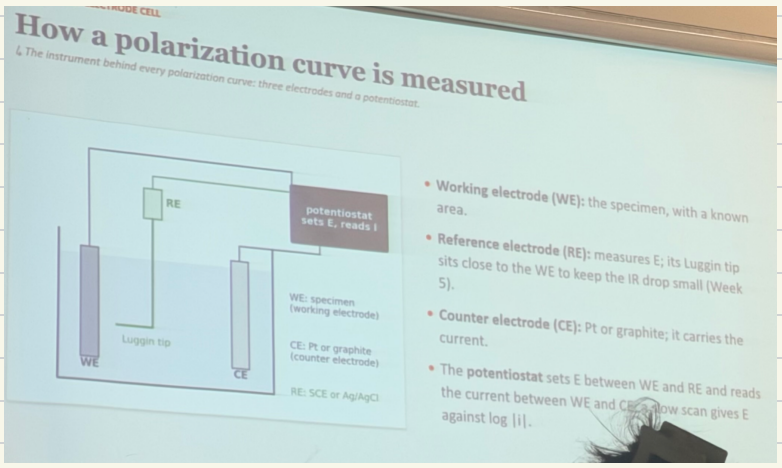
η > 0 ANODIC

η < 0 CATHODIC

(Sağa yakın olandan sola yakın olanı çıkarttım.)





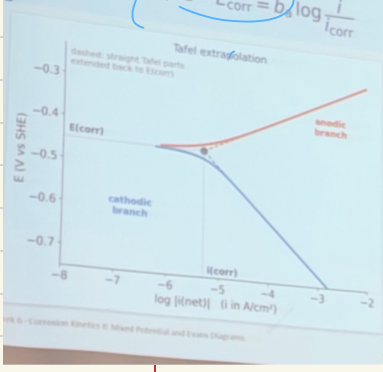


6 · TAFEL EXTRAPOLATION

Extend the straight lines back to E(corr)

↳ The method of this course: use the straight parts of the curve far from E(corr) and extend them back.

$$E - E_{corr} = b_a \log \frac{i}{i_{corr}} \qquad E - E_{corr} = b_c \log \frac{|i|}{i_{corr}}$$



- Measure the curve about ±0.25 V around E(corr).
- Draw straight lines through the **Tafel parts** (at least 50-100 mV away from E(corr)).
- Extend them back to **E(corr)**: they meet at **log i(corr)**.
- The slopes give **b(a)** and **b(c)** at the same time.

EXAMPLE 2 · QUESTION

From i(corr) to mm/y

↳ Example 2 turns today's two corrosion currents into millimetres per year.

Two corrosion currents of carbon steel were found today: 19 µA/cm² for steel in aerated, stagnant seawater and 5.0 µA/cm² for steel in de-aerated acid. Convert both into engineering units and judge them.

Given

- i(corr) = 19 µA/cm² (seawater)
- i(corr) = 5.0 µA/cm² (acid)
- Fe: M = 55.85 g/mol, z = 2, ρ = 7.86 g/cm³
- F = 96485 C/mol

Find

(a) CR in mm/y and mpy for 19 µA/cm²

(b) the mass loss in g m⁻² d⁻¹ for 19 µA/cm²

Week 6 · Corrosion Kinetics II: Mixed Potential and Evans Diagrams

7 · CORROSION RATE UNITS

Penetration units are preferred

↳ First: in which units do engineers express a corrosion rate?

Unit	Meaning	1 mm/y of steel (ρ = 7.86 g/cm³) is
mm/y	millimetres of thickness lost per year (SI)	1 mm/y
mpy	mils per year; 1 mil = 0.001 inch = 0.0254 mm	39.4 mpy
g m⁻² d⁻¹	grams lost per square metre per day	21.5 g m⁻² d⁻¹
mdd	milligrams per square decimetre per day	215 mdd

Penetration units (mm/y, mpy) tell directly how fast a wall gets thinner; mass units need the density.

a)

$$CR(mm/y) = \frac{3{,}27 \times 10^{-3} \cdot (I_{corr} \cdot M)}{z \cdot \rho}$$

$$\frac{3{,}27 \times 10^{-3} \times (19 \times 55{,}85)}{2 \cdot 7{,}86} = 0.22 \text{ mm/y}$$

$$= 8.7 \text{ mpy}$$

(1 mm/y = 39.4 mpy)

b)

$$\frac{\text{mass loss}}{\frac{19 \times 10^{-6} \times 55.85}{2 \times 96485} \times 10^4 \times 86400} = 4.8 \text{ g m}^{-2} \text{d}^{-1}$$

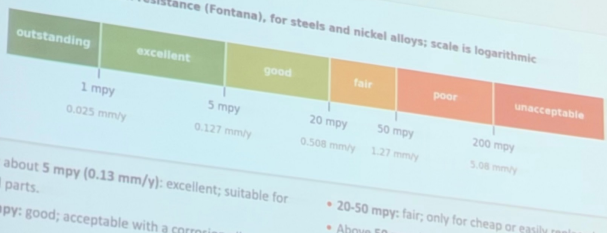
unprotected steel in seawater loses about 2mm in 10 years.

7 · HOW FAST IS TOO FAST?

The relative corrosion resistance scale

↳ A number alone means little: a scale tells whether a rate is acceptable.

Relative corrosion resistance (Fontana), for steels and nickel alloys; scale is logarithmic



- Below about 5 mpy (0.13 mm/y): excellent; suitable for critical parts.
- 5-20 mpy: good; acceptable with a corrosion allowance.
- 20-50 mpy: fair; only for cheap or easily replaced parts.
- Above 50 mpy (1.3 mm/y): poor or unacceptable.

EXAMPLE 2 · SOLUTION

From i(corr) to mm/y

$$CR\ (mm/y) = 3.27 \times 10^{-3} \frac{i_{corr}\ (\mu A/cm^2)\ M}{z\rho} \qquad \frac{\Delta m}{At} = \frac{i_{corr} M}{zF}$$

↳ Same formula twice; for the mass loss keep M and leave out ρ.

Data: Fe: M = 55.85 g/mol, z = 2, ρ = 7.86 g/cm³; 1 m² = 10⁴ cm²; 1 d = 86 400 s.

1 mm/y = 39.4 mpy

(a) seawater, 19 µA/cm²

$$CR = 3.27 \times 10^{-3} \times \frac{19 \times 55.85}{2 \times 7.86} = \mathbf{0.22\ mm/y}\ (8.7\ mpy)$$

(b) mass loss per area and day

$$\frac{19 \times 10^{-6} \times 55.85}{2 \times 96485} \times 10^4 \times 86400 = \mathbf{4.8\ g\,m^{-2}\,d^{-1}}$$

Unprotected steel in seawater loses about 2 mm in 10 years.

(a) 0.22 mm/y (8.7 mpy) **(b)** 4.8 g m⁻² d⁻¹

Week 6 · Corrosion Kinetics II: Mixed Potential and Evans Diagrams

AVOID THESE

Common mistakes with Evans diagrams

↳ The errors students make most often with mixed potentials and Evans diagrams.

Mistake	Correct
"E(corr) is the Nernst potential of the metal"	E(corr) is a **mixed** potential, between the two E(eq) values; it comes from kinetics
"The less noble metal always corrodes faster"	The rate depends on the **kinetics of both reactions** (zinc and iron, Example 1)
b(c) used with a plus sign	The cathodic Tafel slope is **negative**: b(a) − b(c) = b(a) + |b(c)|
Current densities added on different areas	In a galvanic couple use **currents** I = i·A
"Better steel → less corrosion in aerated water"	Under O_2 diffusion control i(corr) = i(L): the **water** sets the rate
"At E(corr) the current is zero, so nothing happens"	Only the **net** current is zero; the metal dissolves at i(corr)
Tafel lines drawn through points near E(corr)	Only the straight parts (≥ 50–100 mV away from E(corr))

Week 6 · Corrosion Kinetics II: Mixed Potential and Evans Diagrams

7 · FROM CURRENT DENSITY TO MM/Y

Faraday's law in engineering units

↳ We have just measured i(corr); Faraday's law turns it into a thinning rate.

$$CR\ (mm/y) = 3.27 \times 10^{-3} \frac{i_{corr}\ (\mu A/cm^2)\ M}{z\rho} \qquad \frac{\Delta m}{At} = \frac{i_{corr} M}{zF}$$

→ mass loss için bu formül

1 **Faraday (Week 5)** $m = \frac{MIt}{zF}$

2 **Divide by the area A** $\frac{m}{At} = \frac{iM}{zF}$ using $I/A = i$

3 **Mass → thickness** $\frac{d}{t} = \frac{iM}{zF\rho}$ using $m = \rho A d$ (d = thickness lost)

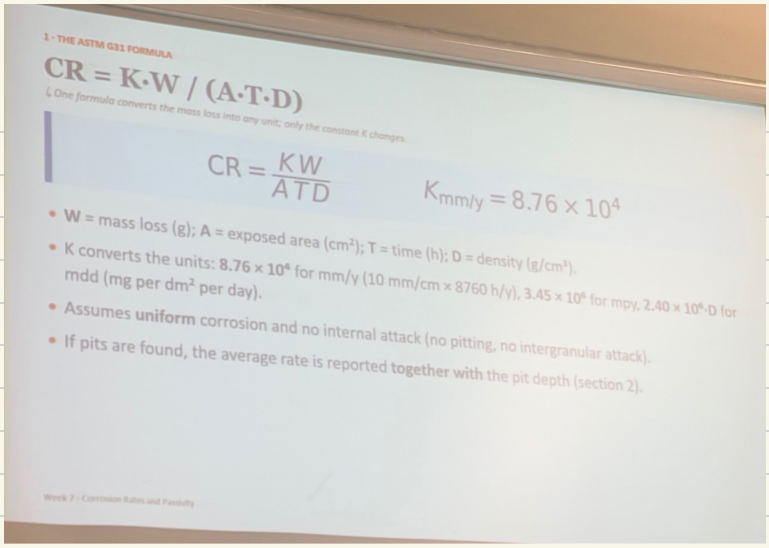
4 **Engineering units** $CR\ (mm/y) = \frac{10 \times 3.15 \times 10^7 \times 10^{-6}}{96\,485} \cdot \frac{iM}{z\rho} = 3.27 \times 10^{-3} \frac{i\ (\mu A/cm^2)\ M}{z\rho}$

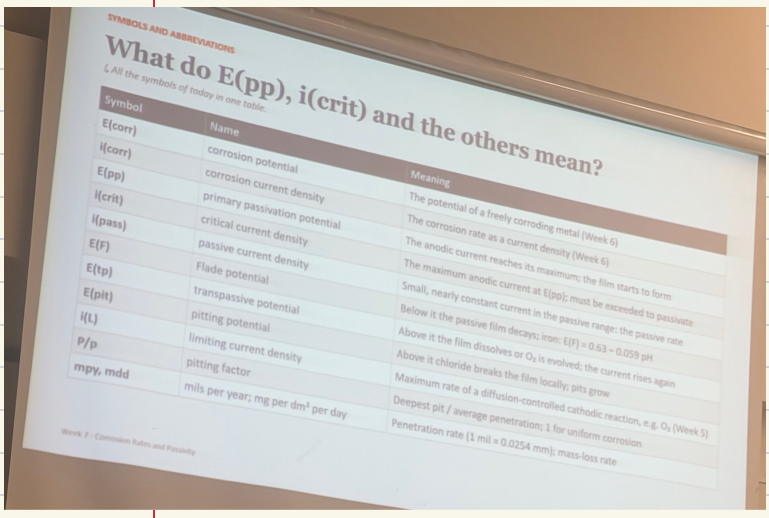
Corossion rate

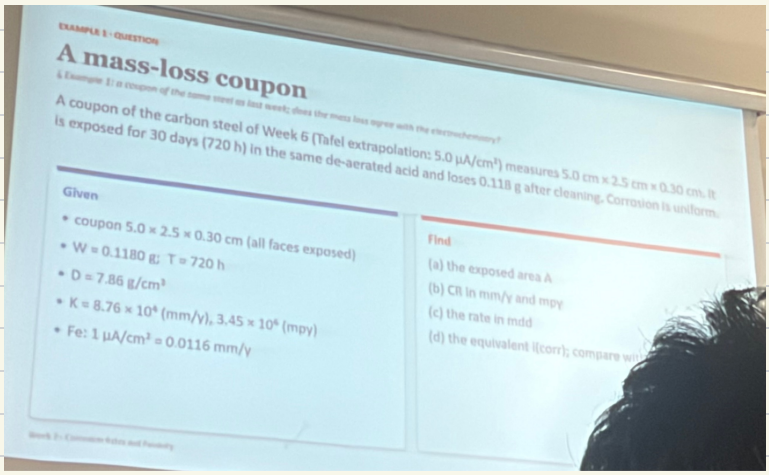
mm per cm × seconds per year × Å per µA, divided by F. F is inside 3.27 × 10⁻³

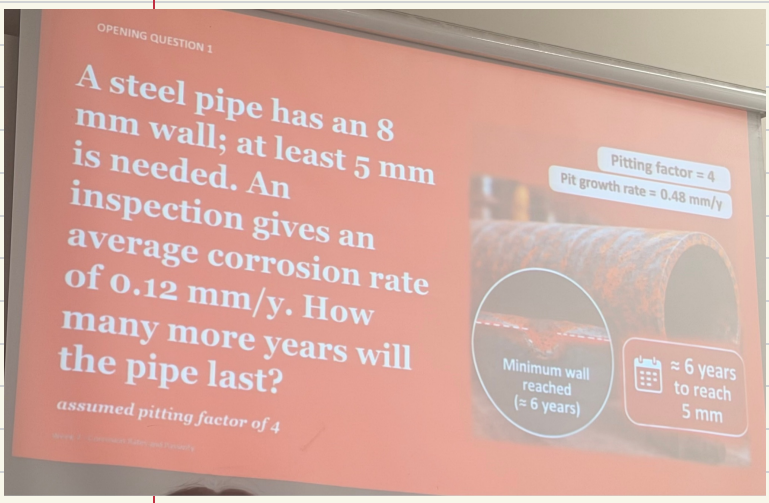
- i(corr) in µA/cm², M in g/mol, z = electrons per atom, ρ in g/cm³.
- 3.27 × 10⁻³ = 10 mm/cm × 3.15 × 10⁷ s/y × 10⁻⁶ A/µA / 96 485 C/mol.
- Iron (M = 55.85, z = 2, ρ = 7.86): 1 µA/cm² ≈ 0.0116 mm/y ≈ 0.46 mpy.
- The second formula gives the mass loss … and time (r = i/zF from We…

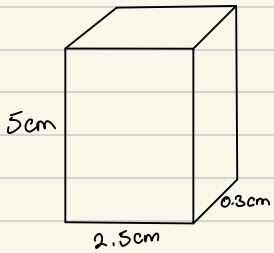
Week 6 · Corrosion Kinetics II: Mixed Potential and Evans Diagrams







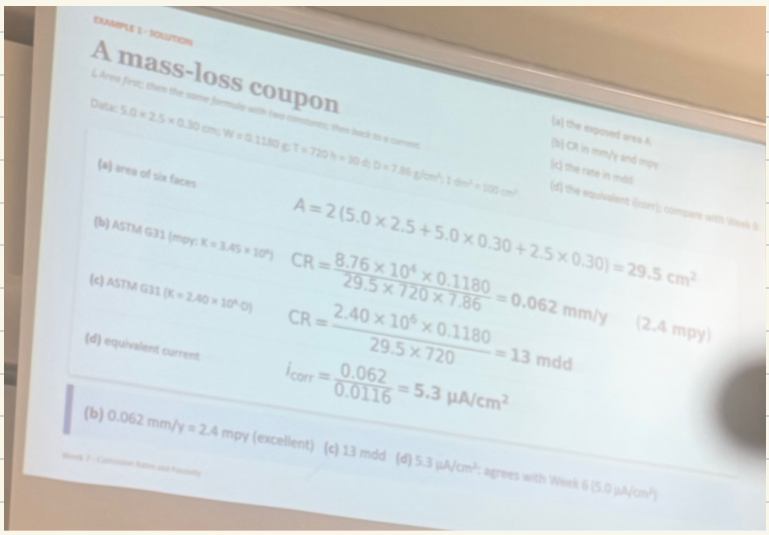




$$CR = \frac{K \times W}{A \times T \times D}$$

$$A = 2(5.0 \times 2.5 + 5.0 \times 0.3 + 2.5 \times 0.3) = 29.5\,cm^2$$

$$CR = \frac{8.76 \times 10^4 \times 0.1180}{29.5 \times 720 \times 7.86} = 0.062\ mm/y\ (2.4\ mpy)$$

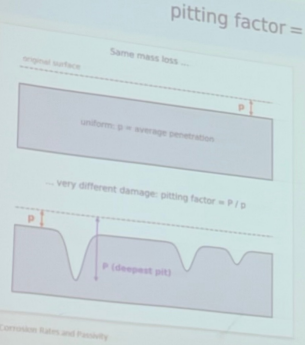


2 · EVALUATING A RATE: PITTING AND SERVICE LIFE

Not the average, the deepest point

The same mass loss can mean very different damage; the average is not enough.

$$\text{pitting factor} = \frac{P}{p} \qquad \text{life} = \frac{t_{wall} - t_{min}}{CR}$$



- p = average penetration (from mass loss); P = depth of the deepest pit.
- **Pitting factor P/p:** 1 for uniform corrosion; 5-20 or more for pitting.
- A wall fails at its thinnest point: use P, not p, for the life.
- Life = corrosion allowance / rate of the deepest attack.

EXAMPLE 2 · QUESTION

How long will the pipe last?

Example 2 answers the opening question.

The steel pipe of the opening question has an 8 mm wall; at least 5 mm is required. Mass-loss coupons give an average rate of 0.12 mm/y. A cut-out section shows a pitting factor of 4.

Given

- wall = 8 mm; minimum = 5 mm
- average rate (mass loss) = 0.12 mm/y
- pitting factor P/p = 4

Find

(a) the corrosion allowance
(b) the life *if* corrosion were uniform
(c) the life with the pitting factor
(d) what should the engineer report?

EXAMPLE 2 · SOLUTION

How long will the pipe last?

Allowance divided by rate; the rate that matters is the rate of the deepest pit.

Data: wall 8 mm; minimum 5 mm; average rate 0.12 mm/y; pitting factor 4.

(a) the corrosion allowance
(b) the life *if* corrosion were uniform
(c) the life with the pitting factor
(d) what should the engineer report?

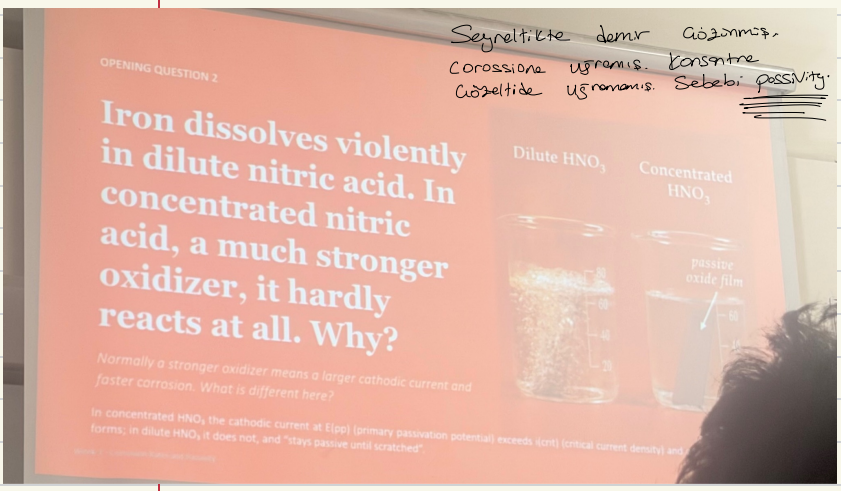
(a), (b) allowance and uniform life: allowance $= 8 - 5 = 3$ mm; uniform: $\frac{3}{0.12} = 25$ **years**

(c) life from the deepest pit: deepest pit: $4 \times 0.12 = 0.48$ mm/y; $\frac{3}{0.48} = 6.3$ **years**

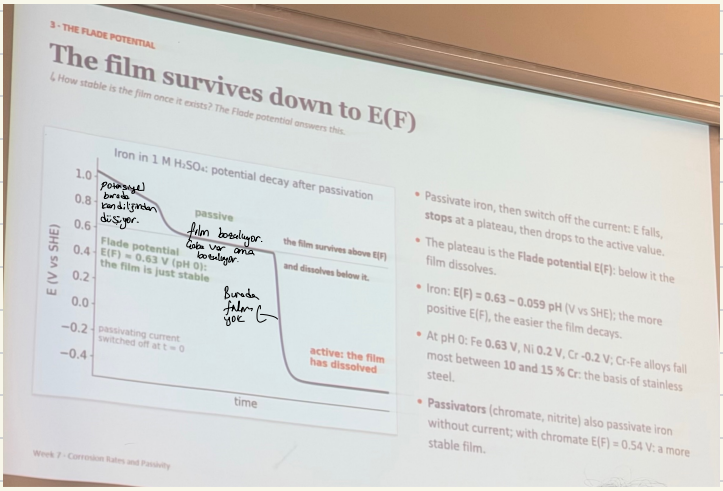
(d) Report about **6 years**, not 25. Inspect the deepest pits regularly, or protect the pipe (coating, cathodic protection...).

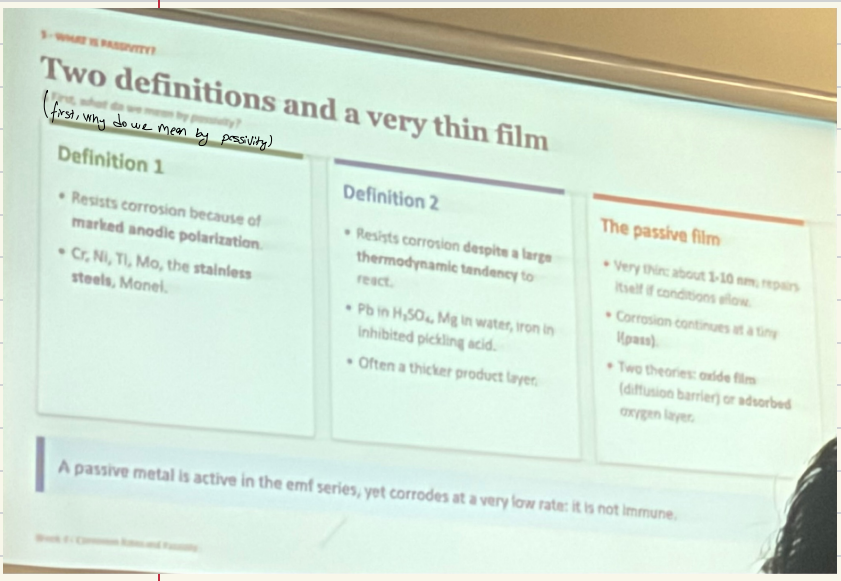
(a) 3 mm **(b)** 25 years if uniform **(c)** about 6 years with pitting factor 4: four times shorter

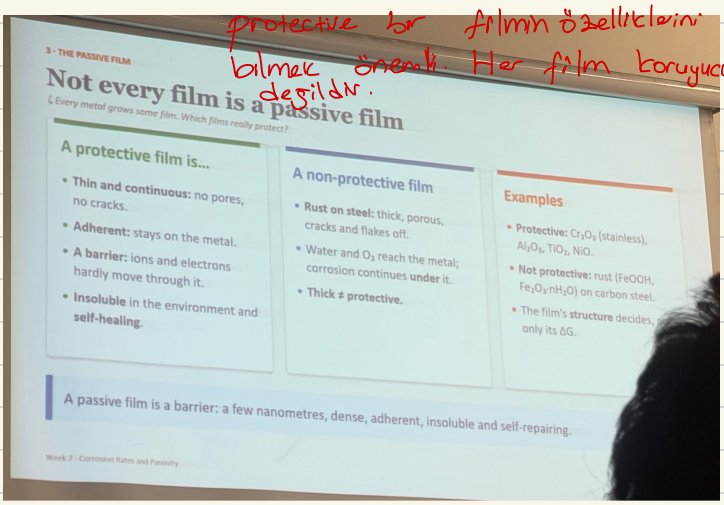
Passivity => Burodan kesin soru gelir. Ya 1. vizede ya da 2. vizede.



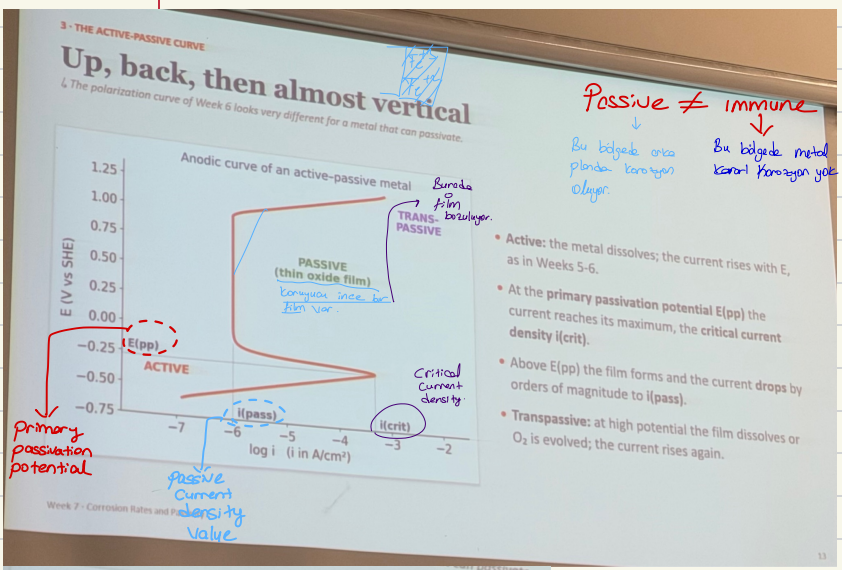
Seyreltikte demir çözünmüş. Corossiona uğramış. Konsantre çözeltide uğramamış. Sebebi: passivity.







Protective bir filmin özelliklerini bilmek önemli. Her film koruyucu değildir.

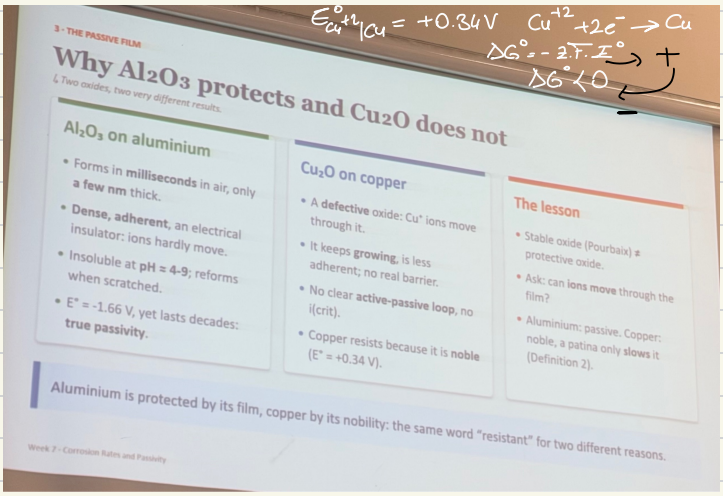


Passive ≠ immune

Bu bölgede orta planda korozyon oluyor.

Bu bölgede metal kararlı korozyon yok.

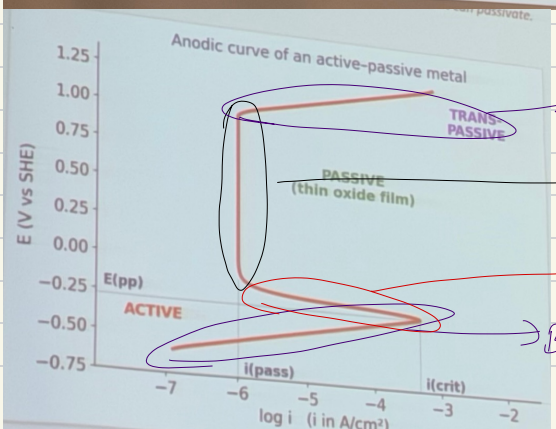
Burada film bozuluyor.



$E^\circ_{Cu^{+2}/Cu} = +0.34V$ $Cu^{+2} + 2e^- \rightarrow Cu$

$\Delta G^\circ = -2F.E^\circ \rightarrow +$

$\Delta G^\circ < 0$ −

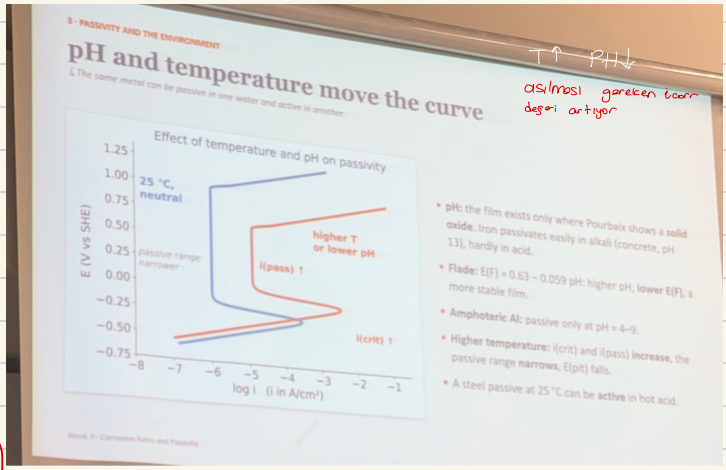


Burada film bozuluyor. Potansiyelle beraber akım da artıyor. (i_{pass}'tan sonra)

Burada potansiyel artıyor ama akım değişmiyor. Film yüzünden. (i_{pass}'te)

Bu bölgede potansiyel artsa bile akım geri geliyor. Sebebi film oluşması. (i_{pass}'a kadar geri geliyor)

Burada akımla beraber potansiyel artıyor



T↑ pH↓ aşınması gereken kodr değeri artıyor

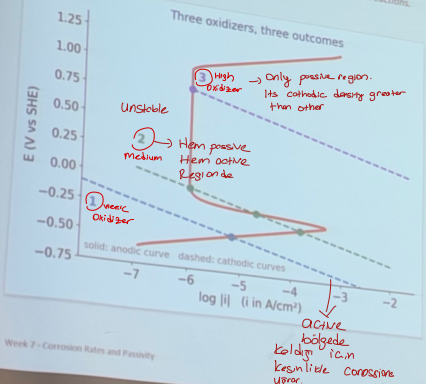
* Kinetik hesaplarda Faraday formülü Nernst denklemi Eq. falan.

* Pure phase diagramı (Suyun) O ve H indirgenme reaksiyonları.

4 - SPONTANEOUS PASSIVATION

The oxidizer decides: active, unstable or passive

↳ Put the active-passive curve on an Evans diagram with different cathodic reactions.



- **Curve 1 (weak oxidizer):** one crossing in the active region: the metal **corrodes**.
- **Curve 2 (medium):** three crossings: passive if the film exists, **active if it is scratched**. Unstable.
- **Curve 3 (strong oxidizer):** cathodic current at E(pp) > **i(crit)**: one crossing in the passive region: **spontaneously passive**.
- Rule: passivation needs a cathodic current larger than **i(crit)** at E(pp).

EXAMPLE 3 - QUESTION

Stainless steel or carbon steel?

↳ Example 3 puts numbers on spontaneous passivation in aerated water.

Two steels are used in aerated, neutral water at 25 °C. The only cathodic reaction is oxygen reduction, which is limited by diffusion: $i(L) = 20\ \mu A/cm^2$ in stagnant water and $80\ \mu A/cm^2$ in flowing water. At E(pp) both steels lie on the plateau of the oxygen curve.

Given

- stainless steel: $i(crit) = 10\ \mu A/cm^2$, $i(pass) = 0.10\ \mu A/cm^2$
- carbon steel: $i(crit) = 100\ \mu A/cm^2$ (no passive film reached)
- $i(L) = 20\ \mu A/cm^2$ (stagnant), $80\ \mu A/cm^2$ (flowing)
- Fe-base: $1\ \mu A/cm^2 \approx 0.0116$ mm/y

Find

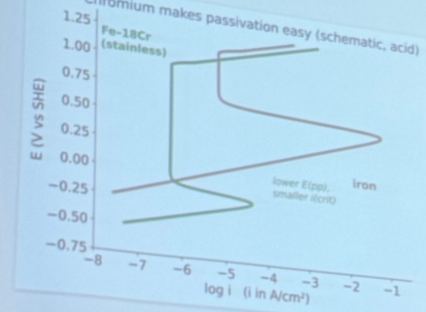
(a) does the stainless steel passivate by itself?
(b) does the carbon steel?
(c) both corrosion rates in stagnant water (mm/y)
(d) what changes in flowing water?

i(L) limit akım sınırlayıcı olacak. Ben ne kadar akım verirsem vereyim. Limit akım beni sınırlayacak. Akım artmadığı için active bölgede kalacam ve corrosiona uğrayacağım.

4 - STAINLESS STEEL

Chromium makes iron easy to passivate

↳ Why is "stainless" steel stainless? Compare its curve with that of iron.



- At least ≈ 11–12 % Cr (≥ 10.5 % by definition) gives a continuous Cr_2O_3-rich film.
- Cr **lowers E(pp)** and **i(crit)** by orders of magnitude and widens the passive range.
- So dissolved O_2 **alone** can passivate it: spontaneous passivation (Example 3).
- **Ni** helps in reducing acids; **Mo** raises E(pit) against chloride.
- The film needs **oxygen** to heal: without O_2 (crevices) stainless can turn active.

Cromun varlığı → i crit düşmüş passive range ↑ Sp ↓

EXAMPLE 3 - SOLUTION

Stainless steel or carbon steel?

↳ Compare what oxygen can supply with what each steel needs.

Data: $i(L) = 20\ \mu A/cm^2$ (stagnant), $80\ \mu A/cm^2$ (flowing); stainless: $i(crit) = 10$, $i(pass) = 0.10\ \mu A/cm^2$; carbon steel: $i(crit) = 100\ \mu A/cm^2$.

(a) stainless steel	stainless : $i_L = 20\ \mu A/cm^2 > i_{crit} = 10\ \mu A/cm^2 \Rightarrow$ **passive**
(b) carbon steel	carbon steel : $i_L = 20\ \mu A/cm^2 < i_{crit} = 100\ \mu A/cm^2 \Rightarrow$ **active**, $i_{corr} = i_L$
(c) corrosion rates (stagnant)	stainless : $0.10\ \mu A/cm^2 \times 0.0116 = 0.0012$ mm/y carbon steel : $20\ \mu A/cm^2 \times 0.0116 = 0.23$ mm/y
(d) flowing water	flowing : $i_L = 80\ \mu A/cm^2 > i_{crit} = 10 \Rightarrow$ stainless still passive carbon steel : $80\ \mu A/cm^2 \times 0.0116 = 0.93$ mm/y

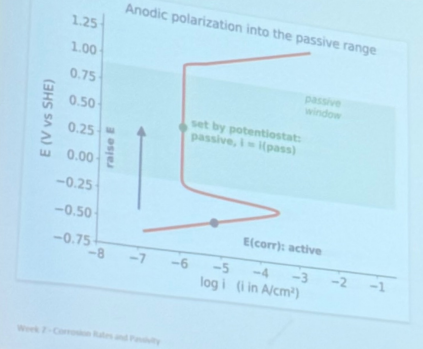
artmasının sebebi difüzyon katsayısı gibi değişebilen etkilerden dolayı olur.

(c) stainless 0.0012 mm/y (passive) against carbon steel 0.23 mm/y: about **200 times** slower

4 - ANODIC PASSIVATION

Making a metal more anodic can protect it

↳ If the oxidizer cannot passivate the metal, a potentiostat can.



- Active at E(corr): the potentiostat **raises** E above E(pp); the current passes i(crit), then drops to **i(pass)**.
- The same without current: a **better cathode**, e.g. Pd or Cu alloyed into stainless steel (cathodic alloying).
- Principle of **anodic protection** (Week 15): steel tanks for H_2SO_4, stainless in acids.
- E must stay in the **passive window:** below E(pp) active, above E(pit) or E(tp) attacked.
- Only for **active-passive** metals: for Cu or Zn, more anodic = faster corrosion.

Cathodic koruma yöntemleri daha çok tercih edilir Denizaltı — Toprakaltı uygulamaları.

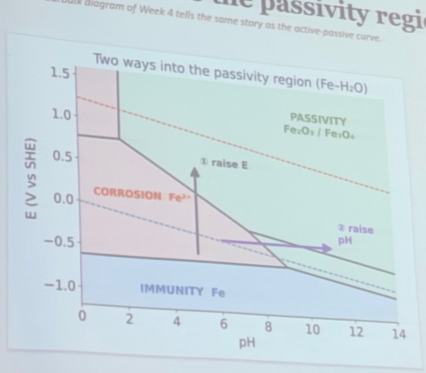
* Cl pitting oluşturuyor. Malzemeyi corrosiona uğratıyor Cl O_2 afinitesi metalden daha fazla

4 - PASSIVITY AND THE POURBAIX DIAGRAM

Two ways into the passivity region

↳ The Pourbaix diagram of Week 4 tells the same story as the active-passive curve.

Ne kadar kararlı olduğunu eğriler söyler.

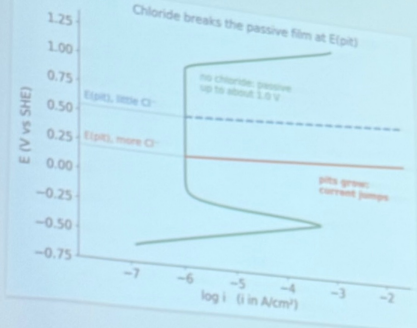


- Pourbaix **corrosion region** ↔ the **active** part of the curve; **passivity region** ↔ the **passive** part.
- ① **Raise E** at constant pH: anodic polarization or a stronger oxidizer (HNO_3, O_2).
- ② **Raise pH** at constant E: alkaline water, concrete (pH 13).
- The boundary Fe^{3+}/Fe_2O_3 is roughly where **E(pp)** lies.
- Pourbaix says the oxide is **stable**, not that it **protects** (Cu_2O, chloride).

3 - CHLORIDE AND THE BREAKDOWN OF PASSIVITY

Above E(pit) the film breaks locally

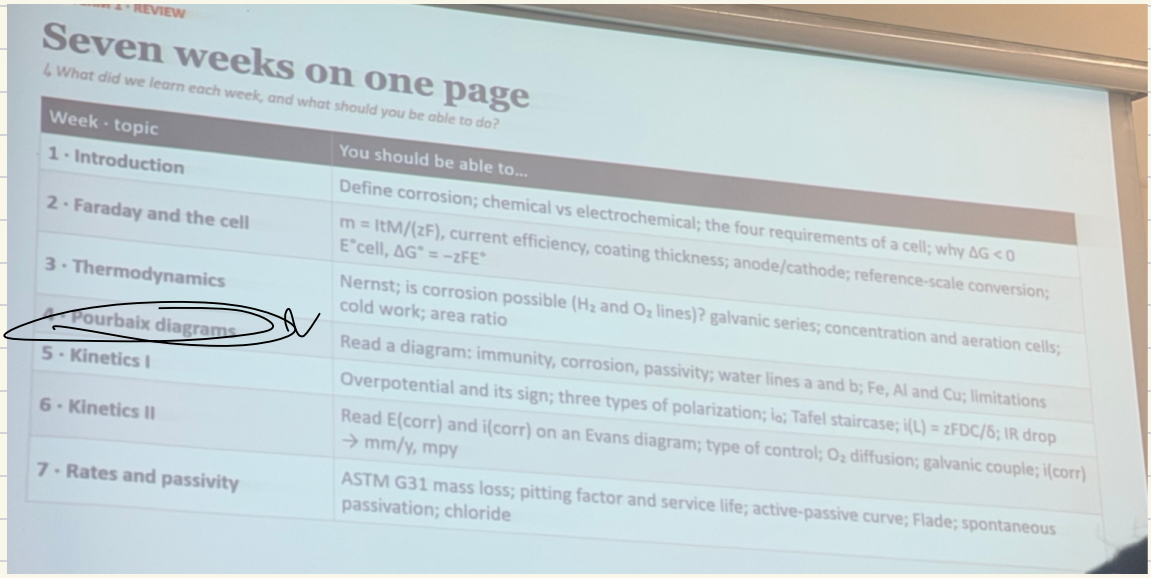
↳ The passive film has a weak point: chloride ions.



- Chloride ions adsorb on the film and break it at **weak spots** (inclusions, scratches).
- Above the **pitting potential E(pit)** the current jumps: small, deep pits grow.
- **More Cl^-** (or higher temperature) → **lower E(pit)**: a smaller passive range.
- Seawater is the classic case: a passive alloy can still be attacked **locally**.
- **Ti, Ta, Mo, Zr** stay passive even in high Cl^-; Fe, Cr and stainless steels lose passivity.

Sınav

* Faraday
* $\Delta G = -zFE$
* Nernst H_2 O_2 lines



REVIEW

Seven weeks on one page

What did we learn each week, and what should you be able to do?

Week · topic	You should be able to…
1 · Introduction	Define corrosion; chemical vs electrochemical; the four requirements of a cell; why $\Delta G < 0$
2 · Faraday and the cell	$m = ItM/(zF)$, current efficiency, coating thickness; anode/cathode; reference-scale conversion; $E°cell$, $\Delta G° = -zFE°$
3 · Thermodynamics	Nernst; is corrosion possible (H_2 and O_2 lines)? galvanic series; concentration and aeration cells; cold work; area ratio
4 · Pourbaix diagrams	Read a diagram: immunity, corrosion, passivity; water lines a and b; Fe, Al and Cu; limitations
5 · Kinetics I	Overpotential and its sign; three types of polarization; i_0; Tafel staircase; $i(L) = zFDC/\delta$; IR drop
6 · Kinetics II	Read E(corr) and i(corr) on an Evans diagram; type of control; O_2 diffusion; galvanic couple; i(corr) → mm/y, mpy
7 · Rates and passivity	ASTM G31 mass loss; pitting factor and service life; active-passive curve; Flade; spontaneous passivation; chloride

* Suyun + reaksiyonu
* Polarizasyon ve türleri nedir.
* Corossion hesabı
* mass loss hesabı
* Active passive hesabı
* PH sıcaklık passivelik etkisi
* Referans hesabı = Sce = SHE + X değeri